

Prdm1 Positively Regulates Liver Group 1 ILCs Cancer Immune Surveillance and Preserves Functional Heterogeneity

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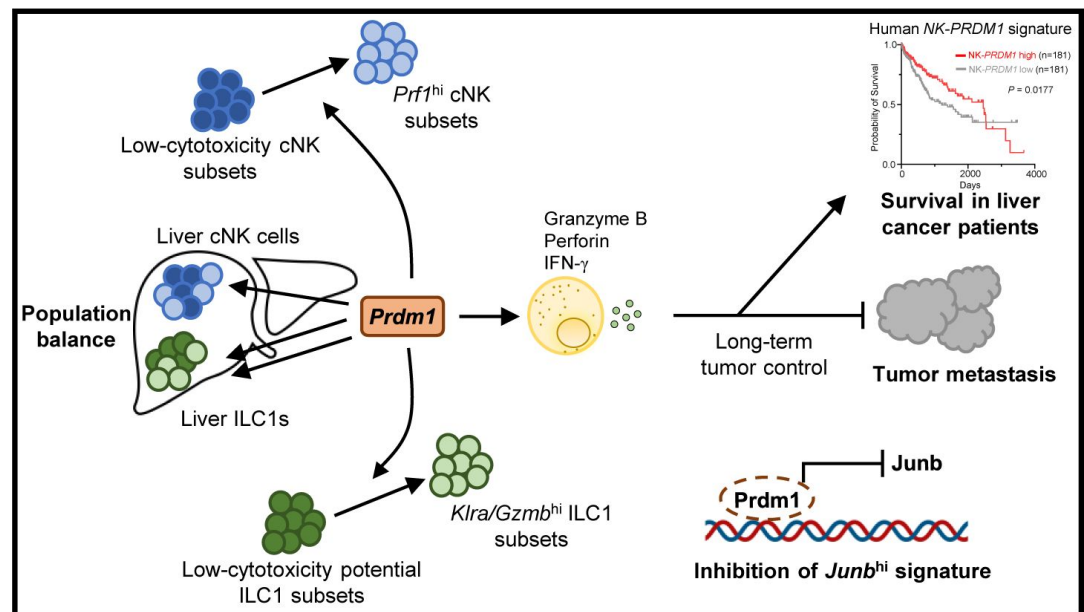
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Abstract

Abstract

Group 1 innate lymphoid cells (ILCs) comprise conventional natural killer (cNK) cells and type 1 innate lymphoid cells (ILC1s). The main functions of liver cNK cells and ILC1s not only include directly killing target cells but also regulating local immune microenvironment of the liver through the secretion of cytokines. Uncovering the intricate mechanisms by which transcriptional factors regulate and influence the functions of liver cNK cells and ILC1s, particularly within the context of liver tumors, presents a significant opportunity to amplify the effectiveness of immunotherapies against liver malignancies. Using *Ncr1*-driven conditional knockout mouse model, our study reveals the regulatory role of *Prdm1* in shaping the composition and maturation of cNK cells. Although *Prdm1* did not affect the killing function of cNK cells in an *in vivo* cytotoxicity model, a significant increase in cancer metastasis was observed in *Prdm1* knockout mice. Interferon- gamma (IFN- γ), granzyme B, and perforin secretion decreased significantly in *Prdm1* deficient cNK cells and liver ILC1s. scRNA sequencing data also provided evidences that *Prdm1* maintains functional subsets of cNK cells and liver ILC1s and facilitates communications between cNK cells, liver ILC1s and macrophages. The present study unveiled a novel regulatory mechanism of *Prdm1* in cNK cells and liver ILC1s, showing promising potential for developing innovative immune therapy strategies against liver cancer.

Graphical abstract



eLife assessment

The authors investigated the requirement and function of Blimp1/*Prdm1* in murine natural killer (NK) cells and the ILC1 lineage of innate lymphoid cells, using a conditional knockout model. The single-cell mRNA-seq data provided here represent a **valuable** resource for the community, but the lack of mechanistic investigations leaves the study partially **incomplete**. The work will be of interest to the fields of innate lymphoid cell biology and tissue immunology.

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Introduction

Group 1 ILCs consist of cNK cells and ILC1s (1, 2), with distinct developmental trajectories and effect molecules (3). Both cNK cells and ILC1s play indispensable roles in combatting viral infections (4), maintaining local immune homeostasis (5), eradicating malignant transformed cells (6), and fostering cross-talk with adaptive immunity (7). cNK cells demonstrate potent cellular cytotoxicity, facilitating direct elimination of target cells (8). On the contrary, a defining attribute of ILC1s is their predominant cytokine-mediated functions, with limited cellular killing capacity (9). In a state of homeostasis, liver group 1 ILCs (CD45⁺CD3⁺NK1.1⁺NKp46⁺) can be discriminated into cNK cells and ILC1s by the differential expression of CD49a and CD49b (2): cNK cells are marked by the expression of CD49b, while liver ILC1s exhibit a distinctive positivity for CD49a. Tumor Necrosis Factor Related Apoptosis Inducing Ligand (TRAIL) is also expressed on liver ILC1s, but not on cNK cells (10). Transcriptional factors (TFs) such as T-bet, Nfil3, PLZF, and ID2 are required for cNK cells and liver ILC1s development or generation of their progenitors (11–14). Eomes is considered to be necessary for NK cell maturation, but not for ILC1s (15). Liver environment facilitated T-bet expression in the early stage of NK cells development, which

results in Eomes repression. The repression of T-bet is required for Eomes⁺ NK cells (16). On the contrary, deficiency of Hobit results in the depletion of ILC1s, while only leaving little impact on liver NK cells (17). These studies suggest that some TFs may, or usually, play different roles in NK cells and ILC1s.

During the development of liver cancers, the functions of immune cells are often inhibited, resulting in the formation of an immunosuppressive tumor microenvironment, and sometimes even systemic immune suppression (18). Transforming a cold tumor into a hot tumor with a stronger immune response is one of the goals of many cancer immunotherapies (19). Research targeting the immune system is showing increasing clinical promise in liver cancer treatment (20). One of our recent studies also found that Toll-like receptor agonists can significantly enhance the anti-tumor effect of Sorafenib by reconstructing the tumor immune microenvironment and reshaping the vascular system (21).

The anti-tumor activity of cNK cells in the liver is relatively well-established, whereas the relationship between liver ILC1 and tumors is still a topic of controversy. Clinical data showed that liver tumors with a higher infiltration of NK cells are associated with a better prognosis (22). Decreased NK cell activity is closely correlated with the malignancy of liver tumors and represents a significant risk factor for recurrence (23). The tumor microenvironment orchestrates the transformation of cNK cells into ILC1s in a TGF- β -dependent manner, resulting in their diminished capacity to control tumor growth and metastasis. This process ultimately promotes tumor immunoevasion (24). However, researches also revealed that, distinct from directly inhibiting tumor growth, the primary function of ILC1 is to suppress the seeding of metastatic tumor cells in liver tissue (6). Comprehensive research in this field is essential to harness the precision of group 1 ILCs targeted immunotherapy for liver cancers.

The transcription factor network governs the function of group 1 ILCs and the balance between cNK and ILC1. Our research team, along with others, has observed that one of the transcription factor in TGF- β pathway, Smad4, promotes the shift in balance from cNK cells towards ILC1 in a TGF-beta-independent pathway, and simultaneously, it positively regulates the expression of another transcription factor, PR domain 1 (Prdm1/Blimp1) (25, 26). Prdm1 plays a crucial role in the differentiation of B cells into plasma cells and the homeostasis of T cells (27–29). The function and regulatory network of Prdm1 in NK cells is distinct from B cells and T cells, for its expression is independent of B-cell lymphoma 6 (Bcl-6) and Interferon Regulatory factor 4 (IRF4) but relies on T-bet. In the study that identified Prdm1 as an essential transcriptional factor for NK cell maturation (30), no significant differences were observed in IFN- γ production or cytotoxicity between Prdm1-deficient and wild-type NK cells. Although Prdm1 expression is dependent on IL-15 in immature NK cells and can be further upregulated by IL-12 or IL-21, it plays a role in downregulating the expression of certain cytokine receptors, such as CD25, consequently diminishing the responsiveness of NK cells to IL-2 (31). Notably, some studies even suggested that Prdm1 suppressed the secretion of IFN- γ in NK cells (32). These findings imply that, although Prdm1 promotes NK cell development, it may function as a negative regulator of their activity. However, direct evidence supporting the impact of Prdm1 on NK cell anti-tumor capabilities is still lacking. Furthermore, there has been no investigation into its influence on the homeostasis of cNK cells and ILC1s in the liver, nor has there been an exploration of the underlying mechanisms.

In the current study, we found that the deletion of *Prdm1* in Ncr1⁺ cells resulted in an imbalance in the homeostasis of liver group 1 ILCs, with a shift towards cNK cells. The data also support the essential role of Prdm1 in the cancer surveillance mediated by cNK cells. While Prdm1 positively regulates genes associated with cellular cytotoxicity, it concurrently exerts inhibitory control over certain positive regulators of NK cell development and functionality, such as JunB. Using single-cell RNA sequencing, we have identified a subset of cNK cells within the liver characterized by elevated JunB expression. These cells exhibit decreased expression of genes associated with

cellular cytotoxicity, and their abundance significantly increases following *Prdm1* knockout. Furthermore, our data also support that *Prdm1* promotes the cross-talk between group 1 ILCs and macrophages.

Results

Prdm1 promotes group 1 ILCs homeostasis and terminal maturation

Examination of 363 liver hepatocellular carcinoma (LIHC) patient samples from The Cancer Genome Atlas (TCGA) revealed a positive correlation between the expression of NK cell-associated genes (33) (*NCR1*, *KLRB1*, *CD160*, *PRF1*, etc.) and *PRDM1* expression (Figure 1A). The patients are ordered from highest to lowest based on the expression of NK-*Prdm1* for survival analysis (Figure 1B). Notably, patients exhibiting higher levels of NK-*PRDM1* expression (above the median) experienced better survival outcomes compared to those with lower levels of NK-*PRDM1* expression (below the median) (Figure 1C). Similar results were also found in skin cutaneous melanoma (SKCM, n=454) and lung adenocarcinoma (LUAD, n=497) patients (Supplemental Figure 1, A-F). Patients within the highest quartile of NK-*PRDM1* signature expression demonstrated enhanced overall survival, a result that achieved statistical significance in LUAD and SKCM patients (Supplemental Figure 1, G-I). These data suggested that *PRDM1* in NK cells might be essential for immune surveillance in solid tumors, including liver cancer, and prompted us to investigate the function and mechanism of *PRDM1* in NK cells and ILC1 within the context of liver cancer.

Ncr1-Cre mice were crossed with *Prdm1^{fl/fl}* mice to generate *Ncr1-cre Prdm1^{fl/fl}* mice, which specifically knockout exons 6-8 of *Prdm1* (34) in NKp46 positive cells (Figure 1D). The mice carrying *Ncr1^{Cre/+} Prdm1^{fl/fl}* were referred to as *Prdm1^{ΔNcr1}* mice, and the mice carrying *Ncr1^{+/+} Prdm1^{fl/fl}* were referred to as *Prdm1^{+/+}* mice or wild-type (WT) mice. To further validate the deletion of *Prdm1* in NKp46⁺ cells of *Prdm1^{ΔNcr1}* mice, CD3⁺NK1.1⁺NKp46⁺ cells were sorted and the expression of *Prdm1* was measured by real-time RT-PCR. The expression of *Prdm1* was almost undetectable in CD3⁺NK1.1⁺NKp46⁺ cells of *Prdm1^{ΔNcr1}* mice, while was similar between total splenocytes in *Prdm1^{+/+}* and *Prdm1^{ΔNcr1}* mice (Figure 1E, E and F) indicating successful, and specific knockout of *Prdm1* in NKp46⁺ cells.

Proportion and absolute number of cNK cells in blood, bone marrow, lung, liver, spleen, and lymph nodes were analyzed by flow cytometry. Compared with *Prdm1^{+/+}* mice, the percentage and absolute number of cNK cells (CD45⁺CD3⁺NK1.1⁺NKp46⁺) among lymphocytes was decreased in all of these tissues, whereas increased number of NK cells were observed in bone marrow (Figure 1G; Supplemental Figure 2A). NK cell terminal maturation can be divided into four stages according to the expression of CD11b and CD27 (35). Stage IV (CD11b⁺CD27⁺) NK cells show higher level of effector molecules than any other stages and are considered as the most mature NK cells (36). The maturation of cNK cells (gated by CD45⁺CD3⁺NK1.1⁺NKp46⁺CD49b⁺) from blood, bone marrow, lung, liver, spleen, and lymph nodes were assessed, based on the expression of CD11b and CD27. Compared with *Prdm1^{+/+}* mice, the proportion of the most mature CD11b⁺CD27⁺ NK cells were significantly decreased in all of the analyzed tissues in *Prdm1^{ΔNcr1}* mice (Figure 1H; Supplemental Figure 2B). Killer cell lectin-like receptor subfamily G member 1 (KLRG1) is a lectin-like receptor which was considered as another marker of NK cell maturation (37). In both *Prdm1^{+/+}* and *Prdm1^{ΔNcr1}* mice, cNK cells from the liver and lung had the highest expression of KLRG1, following by blood and spleen (Supplemental Figure 2C). The lowest KLRG1 expression was observed in cNK cells derived from lymph nodes and bone marrow, indicating the presence of the most immature cNK cells in these tissues. Consistent with CD11b/CD27 based maturation

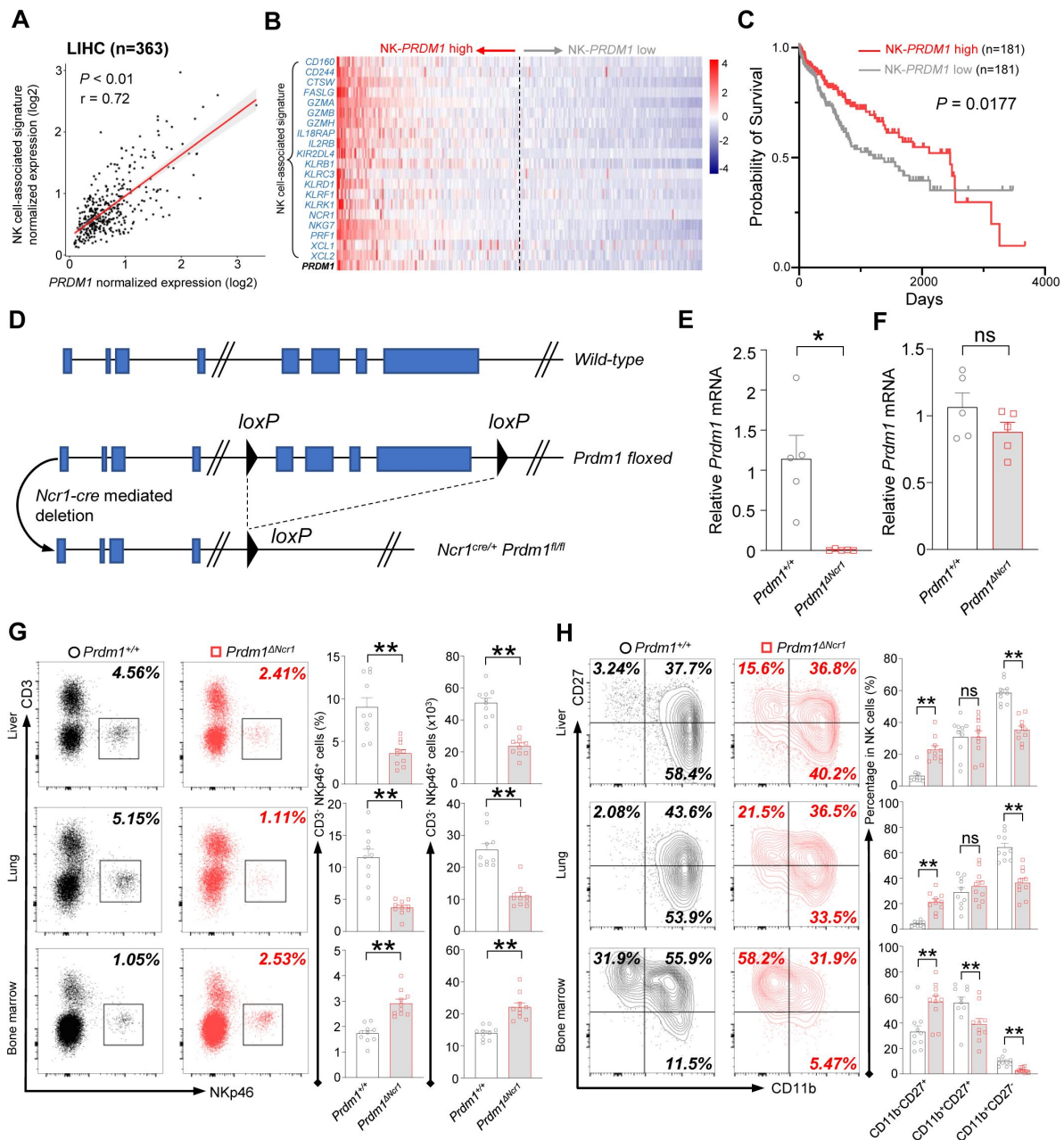


Figure 1.

Prdm1 promotes NK cell homeostasis and terminal maturation. (A) Correlation between the average expression of NK cell-associated signature and *PRDM1* in LIHC (Liver Hepatocellular Carcinoma; n=363) patients sourced from TCGA datasets. (B) Heatmap of the ordered, z-score normalized expression values for *PRDM1* and NK cell-associated genes in liver cancer patients. High and low expression of *NK-PRDM1* signature are indicated. (C) Prognostic value of the *NK-PRDM1* signature for overall survival of liver cancer patients comparing high and low samples with a median cutoff. (D) Schematic representation of the *Prdm1*-conditional knockout mouse model. Targeted exon 6-8 of *Prdm1* (top) is flanked with *loxP* sites (middle). *Ncr1*-expressed Cre recombinase was used to generate the *Prdm1*^{ΔNcr1} allele (bottom). (E and F) Real-time RT-PCR quantification of *Prdm1* expression in NKp46⁺ cells (E) and splenocytes (F) to determine the presence of *Prdm1* (n=5). (G) Representative flow cytometric plots (left) and quantification (right) of the proportion and absolute number of CD45⁺CD3⁺NKp46⁺ cells among lymphocytes in liver, lung, and bone marrow (n=10). (H) Representative flow cytometric plots (left) of the CD11b and CD27 expression within CD3⁺NK1.1⁺NKp46⁺CD49b⁺ cells in liver, lung, and bone marrow (n=10). Right panel showed the percentage of distinct maturation stages of NK cells. Data are presented as the mean±SEM and were analyzed by 2-tailed, paired t-test. Differences were evaluated between littermates. Each circle and square on graphs represents an individual mouse; P, P-value; r, pearson correlation coefficient; *, P<0.05; **, P<0.01, ns, not significant.

analysis, a significant loss of KLRG1⁺ NK cells in cNK cells was observed in *Prdm1*^{ΔNcr1} mice (Supplemental Figure 2C). Together, these data confirmed that *Prdm1* is required for the terminal maturation of cNK cells among various tissues.

***Prdm1* is required for group 1 ILCs to control tumor metastasis**

Two subpopulations of liver group 1 ILCs (gated by CD45⁺CD3⁺NK1.1⁺NKp46⁺) were further analyzed based on the expression of CD49a and CD49b (Figure 2A). Compared with *Prdm1*^{+/+} mice, the *Prdm1*^{ΔNcr1} mice exhibited an increased percentage of cNK cells (CD49a⁺CD49b⁺) and reduced proportion of ILC1s (CD49a⁺CD49b⁻) (Figure 2A). Of note, the absolute number of both cNK cells and ILC1s were decreased in *Prdm1*^{ΔNcr1} mice, with a more robustly reduction in ILC1s (Figure 2A), which underscored the crucial role of *Prdm1* in maintaining the quantity of both liver cNK cells and ILC1s. Expression level of CD49b was slightly upregulated in *Prdm1*^{ΔNcr1} cNK cells and NKp46⁺ cells in the liver and other tissues (Supplemental Figure 2, D and E). Increased CD49a expression was also observed in *Prdm1*^{ΔNcr1} liver ILC1s, while it showed decreased expression in NKp46⁺ cells in the liver, bone marrow, and lymph nodes (Supplemental Figure 2, F and G). These results indicated the essential role of *Prdm1* in maintaining the balance and hemostasis of cNK cells and ILC1s.

The decreased quantity of NK cell homeostasis and maturation of cNK cells due to *Prdm1* loss motivated us to further explore whether deficiency of *Prdm1* impaired NK cell cytotoxicity. A B2M-deficient cell-based *in vivo* cytotoxicity assay was used to evaluate the effect of *Prdm1* on the cytotoxicity of NK cells (38, 39). B2M-deficient cells do not have detectable Major Histocompatibility Complex I (MHC-I) on the cell surface, making them the target of NK cells (40). Healthy NK cells will reject B2M-deficient donor cells efficiently and the elimination was used to quantify the cytotoxicity of NK cells. Although significant impaired homeostasis and maturation of NK cells were observed in *Prdm1*^{ΔNcr1} mice, no significant difference in the *in vivo* cytotoxicity assay were observed between *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice (Supplemental Figure 2, H and I).

Besides their direct cytotoxic capabilities, NK cells' anti-tumor potential is also influenced by additional factors. These include their ability to counteract tumor-induced immune suppression and exhaustion, enhancing their effectiveness against cancer cells. Furthermore, NK cells can secrete cytokines that activate other immune cells, thereby orchestrating a broader immune response for the elimination of tumors. Moreover, the cytotoxicity assay, due to its relatively short duration, might not fully represent the anti-tumor activity of NK cells when continuously exposed to immune inhibitory signals in the tumor microenvironment. Therefore, we initiated an *in vivo* tumor model to further investigate the impact of *Prdm1* on NK cell anti-tumor capability. B16F10 is a melanoma cell line with low expression of MHC-I, which was susceptible to NK cell killing and usually used to evaluate NK cell anti-tumor capacity (41–43). The B16F10 cells were intravenously (for lung metastasis) or intrasplenic (for liver metastasis) administrated in the mice. The melanoma nodes were quantified three (intravenous injection) or two (intrasplenic injection) weeks after tumor inoculation. Compared with *Prdm1*^{+/+} mice, deficiency of *Prdm1* resulted in more metastasis nodules in both lung (~ 2-fold) and liver (~ 4-fold) (Figure 2, B and D). Histological analysis further confirmed the increased frequency of metastasis tumor foci in *Prdm1*^{ΔNcr1} mice (Figure 2, C and E). In agreement with our *in vivo* data, we also observed decreased IFN-γ secretion in *Prdm1*^{ΔNcr1} mice- derived splenic cNK cells, liver cNK cells, and liver ILC1s when stimulated by IL-18 alone or IL-12/IL-18 (Figure 2F; Supplemental Figure 3; Supplemental Figure 12B), which indicated that *Prdm1* is required for full activation of cNK cells and ILC1s in the context of IFN-γ production. These data implying that *Prdm1* is indispensable for NK cell mediated tumor surveillance.

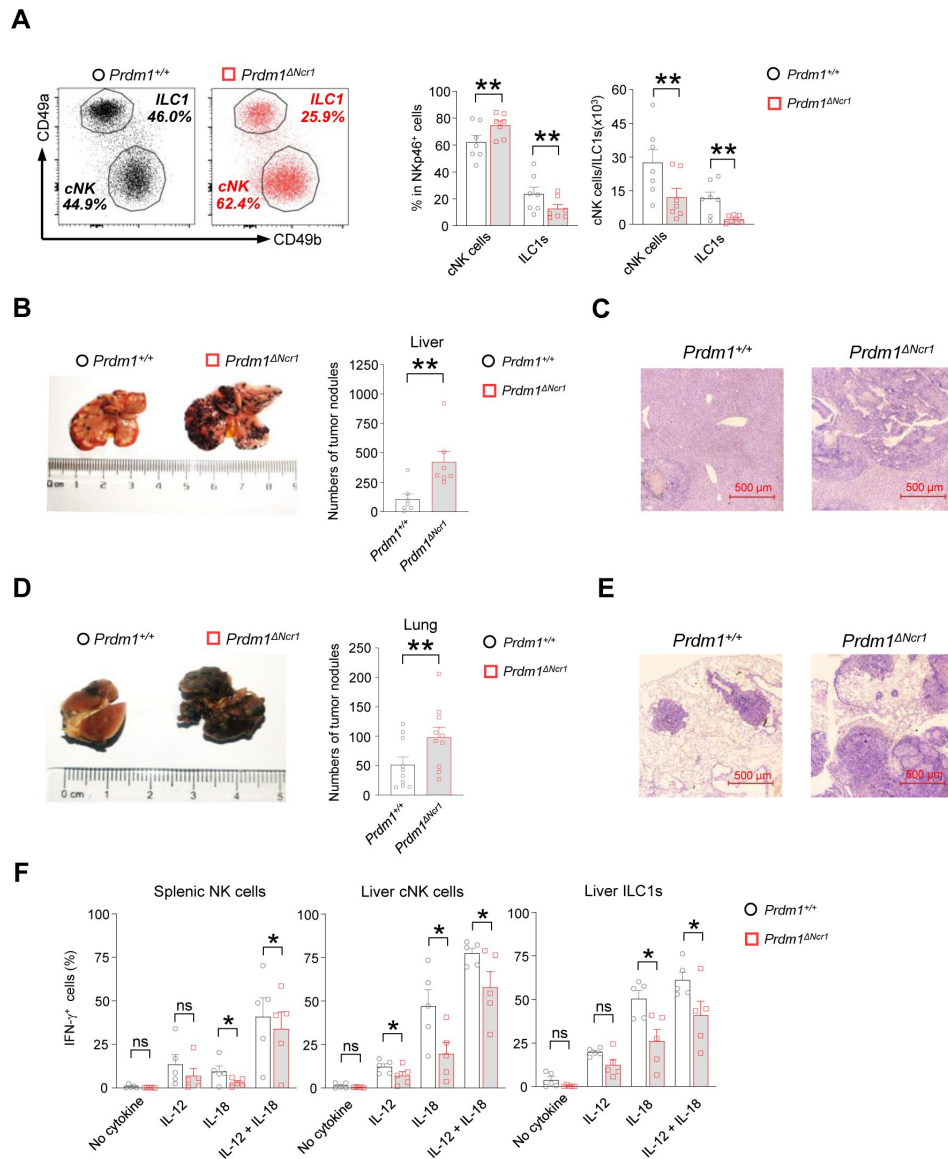


Figure 2.

Prdm1 is required for group 1 ILCs to control tumor metastasis. **(A)** Representative flow cytometric plots (left) of liver cNK cells (CD49a⁺CD49b⁺) and ILC1s (CD49a⁺CD49b⁻) from *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice. The two bar graphs (right) quantitate the percentages and absolute numbers of cells respectively (n=7). **(B and D)** Representative image (left) and quantification (right) of tumor nodes on the livers (n=7) **(B)** and lungs (n=10) **(D)** of *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice at day 14 or 21 after inoculation with B16F10 melanoma cells. **(C and E)** Representative histopathological images of liver **(C)** and lung **(E)** tissues stained by hematoxylin-eosin to detect tumor metastasis. Red bar indicates 500 μm distance under the microscope. **(F)** Liver cells and splenocytes were co-stimulated in the presence or absence of IL-12 and IL-18 for 12 hours. GolgiStop was added 4 hours before intracellular staining of IFN-γ. The graphs showed percentage of IFN-γ⁺ splenic cNK cells, liver cNK cells and ILC1s from *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice (n=5). Data are presented as the mean±SEM and were analyzed by 2-tailed, paired t-test. Differences were evaluated between littermates. Each circle and square on graphs represents an individual mouse; P, P-value; *, P<0.05; **, P<0.01, ns, not significant.

Bulk RNA-seq depicts *Prdm1*-mediated functions in cNK cells

Bulk RNA sequencing of splenic cNK cells ($CD3^+NK1.1^+NKp46^+$) was conducted to uncover the molecular mechanisms by which *Prdm1* regulates NK cell anti-tumor immunity (Figure 3A). Differentially expressed genes (DEGs) between *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice were determined using a criterion of $\log_2$ (fold change) > 0.5 and $P < 0.05$. 445 DEGs were identified out of 17434 protein-coding genes, which consisted of 223 upregulated genes and 222 downregulated genes (Figure 3B).

Gene Ontology (GO) analysis revealed the enrichment of glucuronate metabolism and lymphocyte differentiation in upregulated genes in *Prdm1*^{ΔNcr1} mice derived NK cells (Figure 3C), both of which were associated with cellular growth and development. In contrast, leukocyte mediated cytotoxicity, immune receptor activity, and integrin binding was enriched in the genes which decreased their expression level in *Prdm1*^{ΔNcr1} mice (Figure 3C).

Gene Set Enrichment Analysis (GSEA) showed that NF-kappa B signaling pathway enriched in *Prdm1*-deficient cNK cells (Figure 3D), suggesting the potential targets by *Prdm1* to regulate NK cell function. Increased expression of multiple TFs such as *Junb*, *Batf3*, *Nfkb1*, *Tcf7*, and *Nr4a2* was observed in *Prdm1* knockout cNK cells, suggesting they might be suppressed by *Prdm1* (Figure 3E). Downregulation of granzyme B (*Gzmb*), Perforin (*Prf1*) were observed in *Prdm1* deficient NK cells (Figure 3E), implied decreased anti-tumor ability, which was consistent with increased melanoma metastasis in *Prdm1*^{ΔNcr1} mice (Figure 2, B and D). CXCR6 and CX3CR1 was considered to play an important role in promoting the egress of NK cells from bone marrow. Decreased expression of *Cxcr6* and *Cx3cr1* in *Prdm1*^{ΔNcr1} NK (Figure 3E) might be the reason for the increased quantity of NK cells in bone marrow and decreased number in peripheral tissues (Figure 1G). As a result of the reduced expression levels of *Cxcr6* and *Cx3cr1*, NK cells may not be able to egress from the bone marrow and accumulated therein. Consistent with decreased production of IFN- γ after stimulated by IL-12/IL-18 (Figure 2F), decreased expression of *Il18rap* and *Il12rb2* were observed in *Prdm1*^{ΔNcr1} cNK cells (Figure 3E), implying impaired response to cytokine stimulation.

To confirm the result of RNA-sequencing, the expression of fractalkine receptor (CX3CR1), granzyme B and perforin were analyzed by flow cytometry. The percentage of CX3CR1⁺ cNK cells was significantly decreased in multiple tissues of *Prdm1*^{ΔNcr1} mice, while the proportion of CX3CR1⁺ ILC1 was increased in the liver (Figure 3, F and G). Lower GZMB and PRF1 production was observed in *Prdm1*-deficient splenic cNK cells, liver cNK cells and ILC1s (Figure 3, H-K; Supplemental Figure 4, A-I). Notably, the proportion of GZMB⁺ and PRF1⁺ cNK cells was decreased among almost all of the maturation stages of cNK cells (Figure 3, J and K). The relative mean fluorescent intensities (MFIs) of GZMB and PRF1 consistently show a reduction across all developmental stages in *Prdm1*^{ΔNcr1} NK cells (Supplemental Figure 4, H and I). Yet, no statistical difference of PRF1 was found within the CD11b⁺CD27⁺ and CD11b⁺CD27⁺ subsets, likely due to the relatively lower perforin levels in these populations (Supplemental Figure 4I). These findings suggest that *Prdm1* may directly influence cytotoxic molecule in NK cells, rather than impacting their anti-tumor abilities solely by affecting the maturation phenotype of *Prdm1*-deficient NK cells.

scRNA-seq reveals distinct properties of two clusters from liver group I ILCs following *Prdm1* knockout

To further investigate the effect of *Prdm1* in liver cNK cells and ILC1s and the changes in the hepatic immune microenvironment caused by the deficiency of *Prdm1* in group 1 ILCs, single-cell RNA sequencing (scRNA-seq) was performed for liver CD45⁺ cells (Figure 4A; Supplemental Figure 5A). Initial quality control revealed high-quality of cell purity, library assembly, and sequencing (Supplemental Figure 5B). 10978 cells passed the quality criteria and were selected for further analysis (6,161 from *Prdm1*^{+/+} mice, 4,817 from *Prdm1*^{ΔNcr1} mice). Unsupervised clustering

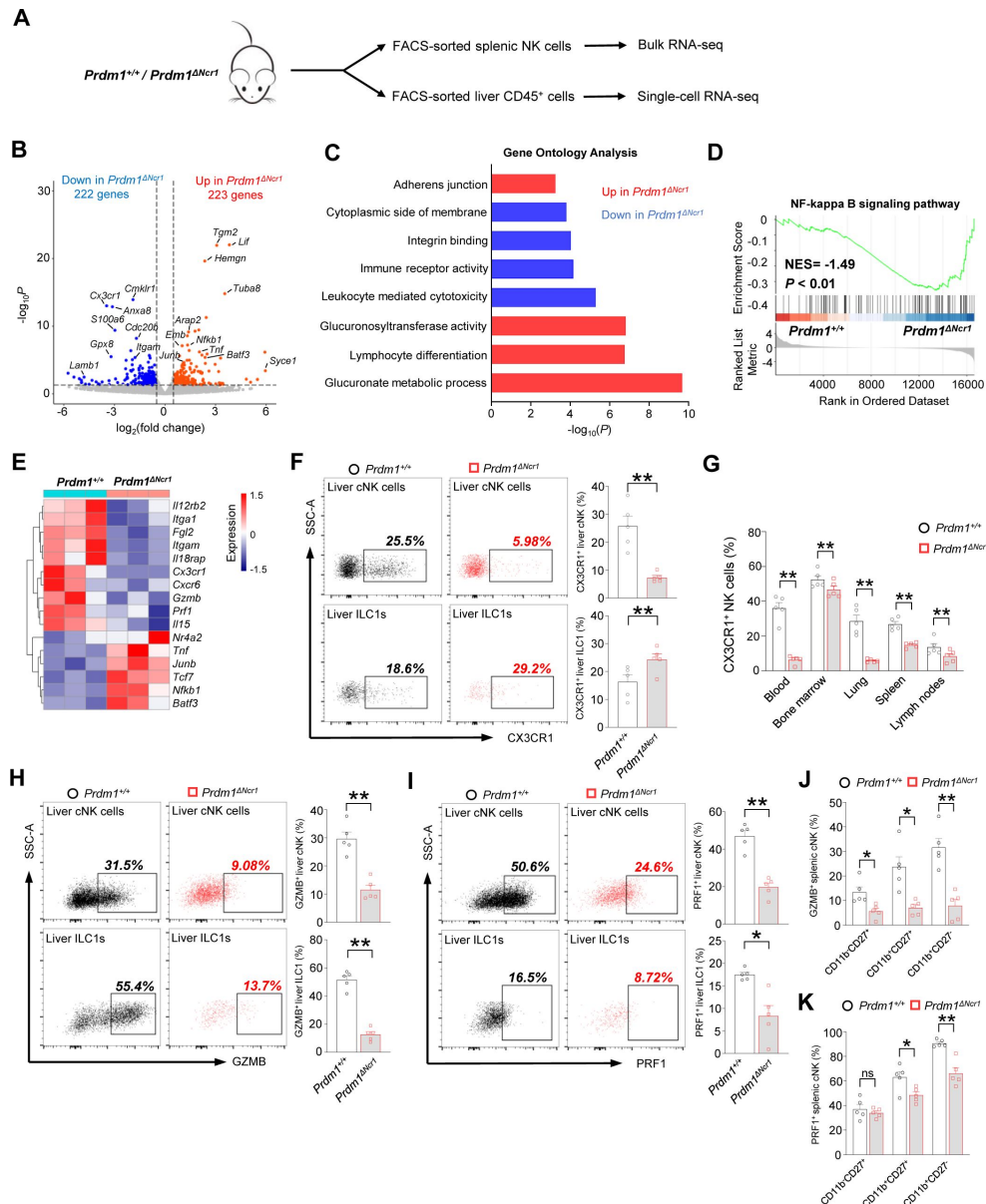


Figure 3.

Bulk RNA-seq reveals *Prdm1*-mediated functions in splenic cNK cells. (A) Splenic cNK cells and liver CD45⁺ cells were sorted from *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice using flow cytometry, and prepared for bulk RNA-seq and single-cell RNA-seq analysis. (B) Volcano plot of the bulk RNA-seq differentially expressed genes (DEGs) ($\log_2$ fold change > 0.5; $P < 0.05$) in splenic cNK cells between *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice. Upregulated and downregulated genes in *Prdm1*^{ΔNcr1} cells compared *Prdm1*^{+/+} cells were highlighted in red and blue. (C) Enriched Gene Ontology (GO) terms of DEGs in *Prdm1*^{ΔNcr1} cells compared *Prdm1*^{+/+} cells. The Enrichment gene set in upregulated (red) and downregulated (blue) genes were indicated in different colour. Bar length represents statistical significance. (D) Gene Set Enrichment Analysis (GSEA) showing the enrichment of NF-kappa B signaling pathway of DEGs in *Prdm1*^{ΔNcr1} cells compared *Prdm1*^{+/+} cells. NES, normalized enrichment score. (E) Heatmap of selected genes from DEGs. Shown is z-score transformed expression of DEGs. (F) Representative flow cytometric plots (left) and cumulative data (right) of the percentage and absolute numbers of CX3CR1⁺ cells in liver cNK cells and ILC1s (n=5). (G) Quantification of CX3CR1⁺ cells in NK cells in blood, bone marrow, lung, liver, spleen, and lymph nodes (n=5). (H and I) Representative flow cytometric plots (left) and cumulative data (right) showing the proportion of GZMB⁺ (H) and PRF1⁺ (I) liver cNK cells and ILC1s from *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice (n=5). (J and K) Proportion of GZMB⁺ (J) and PRF1⁺ (K) splenic NK cells at different maturation stages was analyzed by flow cytometry (n=5). Data are presented as the mean ± SEM and were analyzed by 2-tailed, paired t-test. Differences were evaluated between littermates. Each circle and square on graphs represents an individual mouse; P, P-value; *, $P < 0.05$; **, $P < 0.01$, ns, not significant.

of all sequenced cells based on transcript signatures identified twelve distinct clusters (Supplemental Figure 5C), including B cells, epithelial cells (ECs), CD4⁺ T cells, CD8⁺ T cells, NKT cells, ILC1s, cNK cells, dendritic cells (DCs), monocyte-derived macrophages (MDMs), Monocytes, Kupffer cells (KCs), Neutrophils, and a small number of undefined cells (**Figure 4A**). In *Prdm1*^{ΔNcr1} mice, an increased proportion of MDMs, CD4⁺ T cells, CD8⁺ T cells, Monocytes, and DCs was observed, alongside a decreased proportion of ECs, cNK cells, ILC1s, and NKT cells (Supplemental Figure 5D). Liver cNK cells and ILC1s were identified and discriminated based on the expression of surface markers and distinctive TFs (Supplemental Figure 5, E and F). Compared with other clusters, cNK cells and ILC1s highly expressed *Ncr1* and *Klrb1c* (NK1.1) (Supplemental Figure 5E). cNK cells expressed high levels of *Itga2* (CD49b) and *Eomes*, while ILC1s had high levels expression of *Itga1* (CD49a) and *Tnfrsf10* (Supplemental Figure 5, F and G). Consistent with our flow cytometry data, both cNK cells and ILC1s have significant reduced proportion in *Prdm1*^{ΔNcr1} mouse (Supplemental Figure 5H). In group 1 ILCs from *Prdm1*^{ΔNcr1} mice, there was an increase in the proportion of cNK cells accompanied by a decrease in ILC1s (**Figure 4**, B and C).

To better understand the specific function of *Prdm1* in liver cNK cells and ILC1s, the two subpopulations of liver group 1 ILCs were further analyzed separately using unsupervised clustering and visualized by Uniform Manifold Approximation and Projection (UMAP) (**Figure 4**, D and E). Based on the cluster specific gene expression signature (Supplemental Figure 6A), the subpopulation of liver cNK cells were referred as “*Prf1*^{hi}”, “*Junb*^{hi}”, and “*Cxcr3*^{hi}” cNK cells (**Figure 4D**), with different distribution in *Prdm1*^{ΔNcr1} and *Prdm1*^{+/+} genotype (**Figure 4F**; Supplemental Figure 6B).

The *Prf1*^{hi} cNK cell cluster was defined by high expression of cytotoxicity-related genes, including *Ncr1*, *Gzma*, *Gzmb*, *Prf1*, and *Fgl2* (**Figure 4G**; Supplemental Figure 6, C and D), indicating the strong target-killing ability of this cluster. Although this cluster is present in both *Prdm1*^{ΔNcr1} and *Prdm1*^{+/+} mice, there is a significant reduction in *Prdm1*^{ΔNcr1} mice (**Figure 4F**), indicating the importance of *Prdm1* in maintaining this group of cells. GO analysis further revealed the enrichment signatures of cytotoxicity, response to virus, and lymphocyte mediated immunity in the genes upregulated in *Prf1*^{hi} cNK cell cluster, further confirming the cytotoxic effects of this cluster (**Figure 4H**). These data underscore the crucial role of *Prdm1* in maintaining NK cells with immune effector functions.

The *Junb*^{hi} liver cNK cell cluster distinguished themselves by higher expression of *Junb* compared to other clusters (**Figure 4G**; Supplemental Figure 6, C and D). The predominant majority (92.98%) of *Junb*^{hi} liver cNK cells are derived from *Prdm1*^{ΔNcr1} mice, with less than ten percent (7.02%) originating from *Prdm1*^{+/+} mice (**Figure 4F**). Many signal transduction elements, gene expression regulator, and transcriptional factors, such as *Nfkb1a*, *Tnfrsf10*, *Nr4a1/2/3*, *Batf3*, *Fos*, *Fosb*, *Tcf7*, and *Kit* were upregulated in the *Junb*^{hi} liver cNK cells (**Figure 4G**; Supplemental Figure 6D). The expression of cytotoxicity related genes, such as *Gzmb* and *Prf1*, in *Junb*^{hi} cluster was also lower than other cNK cell clusters (**Figure 4G**). GO analysis showed that the genes upregulated in *Junb*^{hi} liver cNK cells enriched in cell differentiation, cell activation, and transcriptional regulation (**Figure 4H**). GSEA indicates that the NF-kappa B, IL-17, MAPK, and TNF signaling pathways were upregulated in this clusters (**Figure 4**, I-L). GSEA also showed that mitochondrial related pathways, such as mitochondrial protein, oxidative phosphorylation, and respiratory electron transport chain were suppressed in *Junb*^{hi} cNK cell cluster (Supplemental Figure 6, E-G). Increased proportion of *Junb*^{hi} cluster in *Prdm1* deficient cNK cells suggested impaired anti-tumor activity, which was consistent with more melanoma metastasis in *Prdm1*^{ΔNcr1} mice and lower expression of cytotoxicity-related genes in splenic cNK cells based on bulk RNA-sequencing.

The *Cxcr3*^{hi} cNK cell cluster was characterized with high expression of *Cxcr3*, *Ccr2*, and some genes encoding ribosomal subunits such as *Rps7* (Supplemental Figure 6A). Expression of tissue-resident markers Cd69 was also highly expressed in this clusters (Supplemental Figure 6D). The

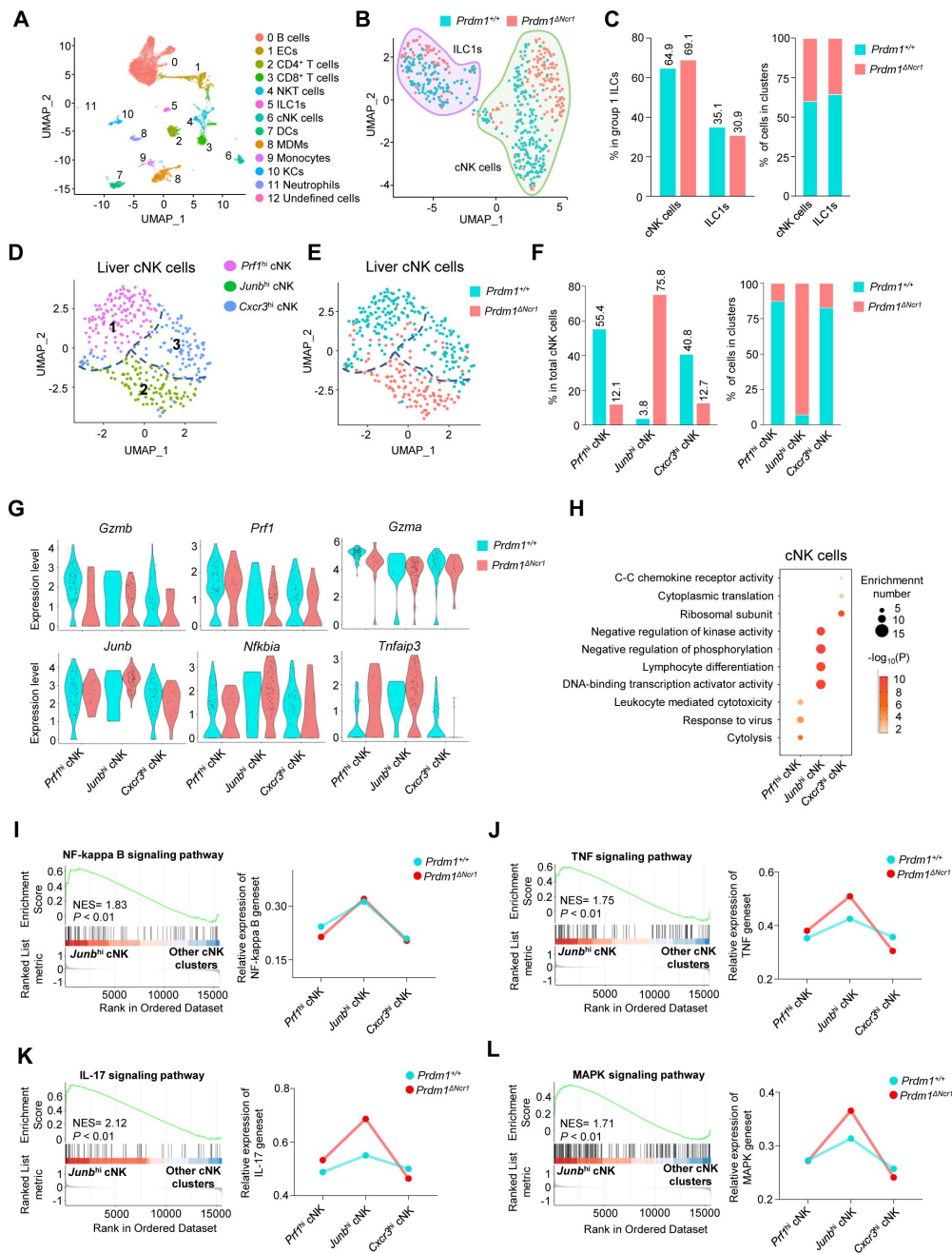


Figure 4.

Different properties of cNK clusters following *Prdm1* knockout. **(A)** Uniform manifold approximation and projection (UMAP) visualization of liver CD45⁺ cells from *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice. Twelve clusters were defined and indicated by distinct colours. Each dot represents a single cell. **(B)** UMAP visualization of liver cNK and ILC1 clusters. Cells were colored by genotypes (*Prdm1*^{+/+}-blue; *Prdm1*^{ΔNcr1}-red). **(C)** Percentages of cNK cells and ILC1s in total group 1 ILCs (left), and their distribution in each cluster (right). **(D and E)** UMAP visualization of three different liver cNK clusters from two mouse strains. **(F)** Proportions of cNK cells among total cNK cells (left; 211 cells in *Prdm1*^{+/+}, and 141 cells in *Prdm1*^{ΔNcr1}) and within clusters (right). **(G)** Violin plots showing the normalized expression of select genes in different cNK clusters. **(H)** Enriched GO term of marker genes in three cNK clusters. Dot size represents enriched gene number, and color intensity represents significance. **(I-L)** GSEA plots (left) depicting the enrichment of NF-kappa B (I), TNF (J), IL-17 (K), and MAPK (L) signaling pathway in *Junb*^{hi} cNK cluster compared with clusters of *Prf1*^{hi} and *Cxcr3*^{hi} cNK cells. Right panel showed dynamic relative expression of the given gene sets from cluster1 to cluster3 between *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1}. Dots represent the average expression of given gene set in each cell, which was calculated through the sum of normalized expression of each individual gene within the designated gene set in every single cell. NES, normalized enrichment score.

enrichment of chemokine receptors in the genes upregulated in the *Cxcr3*^{hi} cluster implying a greater likelihood of this cluster being tissue-resident compared with other cNK cell clusters (Figure 4H). To further confirmed tissue-resident properties of this clusters, we calculated the module score based on top30 DEGs in ILC1 versus cNK clusters, including *Cxcr6*, *Itga1*, *Cd160*, *Cd226*, etc. *Cxcr3*^{hi} cNK clusters have the highest score among all cNK clusters (Supplemental Figure 6H), indicating the similarity with liver ILC1s. In the tumor microenvironment, reports indicated that NK cells could transform into ILC1s (24). If this conversion of cNK cells into ILC1s also occurred under normal physiological conditions, then *Cxcr3*^{hi} cNK cell cluster might be the most susceptible to such transformation.

The significant enrichment of ribosomal subunits and cytoplasmic translation in *Cxcr3*^{hi} cluster (Figure 4H) implied their distinct and active metabolic profile and the capability to mount immune responses. The remarkably decreased proportion of *Cxcr3*^{hi} cNK cell cluster in *Prdm1*^{ΔNcr1} mice (Figure 4F), which emphasized the critical role of *Prdm1* in maintaining this cluster of liver cNK cells, consistent with the flow cytometry result that showed an increase in the number of NK cells in the bone marrow and a decrease in NK cells in peripheral tissues (Figure 1G). To confirm the regulatory effect of *Prdm1* on the migration of NK cells and ILC1 through chemokine receptors, and to validate the scRNA-seq data, we further evaluated the expression of CX3CR1 by flow cytometry. A decrease in the proportion of CX3CR1⁺ NK cells was observed across all tested organs following *Prdm1* knockout. However, it is noteworthy that, in contrast to the trend in NK cells, CX3CR1 expression was increased in liver ILC1s after *Prdm1* knockout (Figure 3, F and G). This not only substantiates the involvement of *Prdm1* in managing NK cell migration but also underscores its distinctive regulatory impacts on the chemokine receptor expressions within NK cells and ILC1 populations. Bulk RNA sequencing data also found that the expression of *Cx3cr1* and *Cxcr6* decreased in *Prdm1*^{ΔNcr1} cNK cells (Figure 3E). These findings supported the hypothesis that *Prdm1*, through regulating chemokine receptor expression levels, influenced the distribution of NK cells in the bone marrow and peripheral tissues, particularly within the liver tissue.

Three clusters of ILC1s were identified from liver ILC1s (Figure 5A), comprising “*Il7r*^{hi}”, “*Klra*^{hi}”, and “*Gzmb*^{hi}” ILC1s. *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} ILC1s seemed to co-cluster largely and have minor difference within the proportion of clusters (Figure 5, B and C; Supplemental Figure 7B). The first two clusters of ILC1s were characterized by higher expression of *Il7r* (CD127) and *Klra5* separately, while the *Gzmb*^{hi} ILC1 cluster was identified by elevated expression of both *Gzma* and *Gzmb* (Figure 5D; Supplemental Figure 7, A, C and D). Additionally, both *Gzma* and *Gzmb* expression were downregulated, and *Junb* was upregulated, in *Prdm1*^{ΔNcr1} mice derived cNK cells and ILC1s compared to those from *Prdm1*^{+/+} mice (Figure 5E). The *Il7r*^{hi} ILC1s cluster was distinguishable from other ILC1 clusters by its high expression of *Il7r*, IL18RA (*Il18r1*), and IFN-γ (*Ifng*) (Figure 5B; Supplemental Figure 7C). The high expression of *Il18r1* and *Ifng* in *Il7r*^{hi} ILC1s indicated this cluster of cells was highly responsive to IL-18 (Supplemental Figure 7D). GSEA and GO analysis showed that IL-17, NF-kappa B, TNF, MAPK signaling pathway and T cell differentiation were activated in the *Il7r*^{hi} ILC1 cluster (Figure 5, F-H; Supplemental Figure 7, E-G). Module scores, calculated based on the expression of feature genes within the *Junb*^{hi} cNK cell cluster, revealed a comparable *Junb*^{hi} signature expression pattern within the *Il7r*^{hi} ILC1 cluster (Figure 5I). Several ILC3 signature genes, such as *Rora*, *Tmem176a*, and *Tmem176b* (44), highly expressed in this cluster (Supplemental Figure 7D). Considering the close relationship between IL-17 mediated immunity response and ILC3 (1, 45), it is plausible that *Il7r*^{hi} ILC1 cluster may be attributed, at least in part, to potential plasticity between ILC1 and ILC3 subsets.

The second liver ILC1 cluster, characterized by high expression of Ly49E (*Klra5*) and Ly49G (*Klra7*), was designated as the *Klra*^{hi} ILC1 cluster (Figure 5D; Supplemental Figure 7D). Notably, there was an elevated proportion of *Klra*^{hi} ILC1s in *Prdm1*^{ΔNcr1} ILC1s (39.7%) compared to *Prdm1*^{+/+} ILC1s (28.1%) (Figure 5C). Liver Ly49E⁺ ILC1s have been identified as possessing greater cytotoxic potential and a more robust viral response compared to liver Ly49E⁻ ILC1s

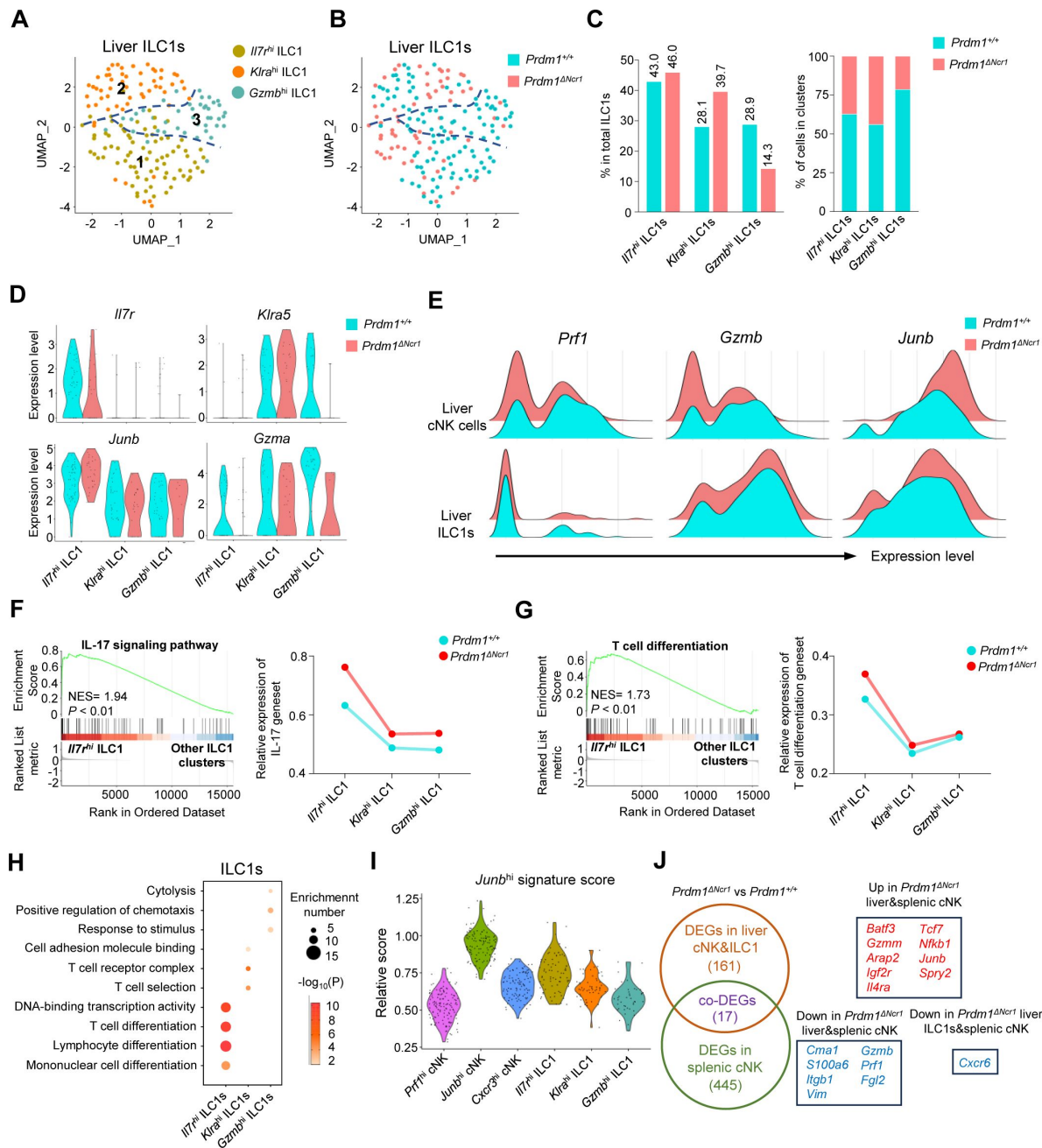


Figure 5.

Different properties of ILC1 clusters following *Prdm1* knockout. (A and B) UMAP visualization of three different liver ILC1 clusters from two mouse strains. (C) Proportions of ILC1s among total ILC1s in different genotypes (left; 114 cells in *Prdm1^{+/+}*, and 63 cells in *Prdm1^{ΔNcr1}*) and within each cluster (right). (D) Violin plots showing the normalized expression of select genes in different ILC1 clusters. (E) Ridge plots showing the normalized expression of *Gzmb*, *Prf1*, and *Junb* in cNK and ILC1 clusters between *Prdm1^{+/+}* and *Prdm1^{ΔNcr1}* cells. (F and G) GSEA plots (left) depicting the enrichment of IL-17 signaling pathway (F) and T cell differentiation (G) in *I17^{hi}* ILC1 cluster compared with clusters of *Klra^{hi}* and *Gzmb^{hi}* ILC1s. Right panel showed dynamic relative expression of the given gene sets from cluster1 to cluster3 between *Prdm1^{+/+}* and *Prdm1^{ΔNcr1}*. Dots represent the average expression of given gene set in each cell, which was calculated through the sum of normalized expression of each individual gene within the designated gene set in every single cell. NES, normalized enrichment score. (H) Enriched GO term of marker genes in three ILC1 clusters. Dot size represents enriched gene number, and color intensity represents significance. (I) Violin plot showing the *Junb^{hi}* signature score for cNK cell and ILC1 clusters, calculated using the signature genes of *Junb^{hi}* cNK cluster. (J) Venn diagram showing overlapping and unique DEGs in comparisons within liver cNK cells, ILC1s and splenic cNK cells between *Prdm1^{+/+}* and *Prdm1^{ΔNcr1}* (left), and 17 overlapped DEGs were shown at the right panel.

(28.1%) (**Figure 5C** [↗](#)). Liver Ly49E⁺ ILC1s have been identified as possessing greater cytotoxic potential and a more robust viral response compared to liver Ly49E⁻ ILC1s ([46](#) [↗](#)). The *Klra*^{hi} cluster exhibited notably high expression of *Ccl5* (Supplemental Figure 7D). Previous research has underscored the pivotal role of CCL5, produced by both cNK cells and ILC1s, in facilitating the accumulation of DCs within the tumor microenvironment, thereby impeding tumor immune evasion, as highlighted in studies ([47](#) [↗](#), [48](#) [↗](#)). The expression of *Ccl5* was reduced in the *Klra*^{hi} cluster of *Prdm1*^{ΔNcr1} ILC1s compared to *Prdm1*^{+/+} ILC1s (Supplemental Figure 7D), which could potentially have a detrimental impact on the ability of *Klra*^{hi} ILC1s to develop a connection between innate and adaptive immune responses.

The *Gzmb*^{hi} ILC1 cluster was identified according to the high expression of *Gzma*, *Gzmb*, and *Fgl2* (Supplemental Figure 7D) compared to other ILC1 clusters. GO analysis also revealed the enrichment in cytolysis and stimulus-response capacity (**Figure 5H** [↗](#)) of the *Gzmb*^{hi} ILC1 cluster. Consistent with the *Prf1*^{hi} cNK cell cluster, the proportion of the *Gzmb*^{hi} cluster among liver ILC1s exhibited a considerable reduction in *Prdm1*^{ΔNcr1} mice compared to *Prdm1*^{+/+} mice, and the expression of *Gzma* also downregulated in *Prdm1*^{ΔNcr1} mice (**Figure 5B** [↗](#)). Previous reports showed that GzmA⁺ ILC1 constituted the main population of liver ILC1s at birth, with the potential target-killing ability ([49](#) [↗](#), [50](#) [↗](#)). Within cNK cells, *Il12rb2*, *Il18r1* and *Il18rap* was highly expressed in *Prf1*^{hi} and *Cxcr3*^{hi} cNK clusters (Supplemental Figure 6I), indicating the IL-18 receptor expression correlated with the NK cell maturation. While in ILC1, these receptors mostly expressed on *Il7r*^{hi} and *Gzmb*^{hi} ILC1 clusters (Supplemental Figure 7D). Significant decreased of *Il18r1* expression in *Prdm1*^{ΔNcr1} cNK cells and ILC1s may associated with the impaired ability to produce IFN-γ.

To investigate the universal transcriptional program between group 1 ILCs across liver and spleen, we have explored DEGs in *Prdm1*^{ΔNcr1} liver cNK cells and ILC1s using our scRNA-seq data. Compared with liver ILC1s, more valuable DEGs was observed in liver cNK cells, including *Junb*, *Kit*, *Tcf7*, *Gzmb*, *Prf1*, etc. (Supplemental Figure 8, A and B). Through the integration of the bulk RNA-seq and scRNA-seq data, we identified 17 DEGs that are regulated by *Prdm1* among liver cNK cells, splenic cNK cells, and liver ILC1s. *Batf3*, *Junb*, *Tcf7*, and *Nfkb1* was upregulated, whereas *Gzmb*, *Prf1*, and *Fgl2* downregulated, in both *Prdm1*^{ΔNcr1} liver and splenic cNK cells (**Figure 5J** [↗](#)). *Cxcr6* was downregulated in liver ILC1s and splenic cNK cells in *Prdm1*^{ΔNcr1} mice (**Figure 5J** [↗](#)). Previous research found that spleen NK cells could be divided into three distinct groups based on their expression levels of CD27, CD62L, CD49a, and CD49b ([51](#) [↗](#)). CD27⁺CD62L⁻ NK cells have remarkable high expression of *Batf3*, while it was only barely expressed in CD27⁺CD62L⁺ and CD27⁻CD62L⁺ NK cells ([51](#) [↗](#)). Based the sequencing data published by Flommersfeld et al., (GSE180978), a notable negative correlation was observed between the expression levels of *Prdm1* and *Batf3* (Supplemental Figure 8I). On top of that, our findings unveiled the negative regulatory influence of *Prdm1* on *Batf3* within both spleen and liver NK cells. This discovery highlights a potential upstream mechanism that may influence the hemostasis of the spleen NK cell subpopulations through *Batf3*.

We also compared the gene expression patterns between *Prdm1* and *Hobit* (homologue of Blimp1) with two published scRNA-seq data ([50](#) [↗](#), [52](#) [↗](#)). Following the knockout of *Hobit*, the DEGs were primarily identified within ILC1s. Conversely, after the knockout of *Prdm1*, a greater number of DEGs were observed in cNK cells. This indicates that *Prdm1* likely possesses a broader range of target genes within cNK cells, whereas *Hobit* appears to have a more pronounced impact on gene expression within ILC1s (Supplemental Figure 8, C-F). There are some overlaps between the downstream transcriptional profile of *Prdm1* and *Hobit* in liver cNK cells and ILC1s (Supplemental Figure 8, G and H), such as *Junb*, *Fosb*, *Tcf7*, *Kit*, *Gzmb*, *Prf1*, and *Cxcr6* was simultaneously upregulated or downregulated in both *Prdm1*^{ΔNcr1} and *Hobit*^{KO} liver cNK cells or ILC1s, indicating the similar regulatory networks of *Prdm1* and *Hobit*.

***Prdm1* facilitates the intercellular communication between liver group 1 ILCs and macrophages**

The reciprocal crosstalk between group 1 ILCs and macrophages plays a critical role in maintaining liver immune homeostasis and anti-cancer immune surveillance (53, 54). The scRNA sequencing analysis identified two well-established subpopulations of liver macrophages: the resident Kupffer Cells (KCs) and the Monocyte-Derived Macrophages (MDMs) (Figure 6, A-C; Supplemental Figure 9A). When comparing the total proportion of macrophages within the immune cell population of the liver between WT and *Prdm1*^{ΔNcr1} mice, there is an increase in *Prdm1*^{ΔNcr1} mice (Figure 6C). To confirm these findings, we utilized flow cytometry to define macrophages, including both KCs and MDMs, gating by CD45⁺Ly6G⁺F4/80⁺CD11b⁺ (Figure 6D). Our analysis showed that, following the deletion of *Prdm1* in Group 1 ILCs, there is a significant increase in both the proportion and number of macrophages in the liver (Figure 6D).

According to the transcriptional profile, liver macrophages further clustered and were labeled as “Ly6c2^{hi}”, “Cxcl2^{hi}”, “Ear2^{hi}” MDMs, and “Mrc1^{hi}”, “C1q^{hi}” KCs (Figure 6A, Supplemental Figure 9, A-E). Increased proportion of MDMs and KCs was observed in *Prdm1*^{ΔNcr1} cells (Supplemental Figure 9B). Within MDMs clusters, Ly6c2^{hi} MDMs mainly compose of *Prdm1*^{+/+} cells, while *Prdm1*^{ΔNcr1} cells concentrated in Cxcl2^{hi} cluster (Figure 6C). The scRNA-seq data reveal that following *Prdm1* knockout in NKp46⁺ cells, there is a decrease in the proportion of KCs within the macrophage population, while the proportion of MDMs increases (Figure 6D). CX3CR1, a chemokine receptor, is extensively utilized to distinguish KCs and MDMs within macrophages. Cells expressing CX3CR1 are identified as MDMs, whereas those without CX3CR1 expression are categorized as KCs (55). Employing flow cytometry and leveraging CX3CR1 expression, we assessed the ratios of KCs and MDMs. However, diverging from the scRNA-seq findings, flow cytometry indicates that post-*Prdm1* knockout in group 1 ILCs, there is a minor increase in the proportion of KCs within the total liver macrophages, and a decrease in the proportion of MDMs (Figure 6D; Supplemental Figure 9B). This discrepancy could stem from the different bases of classification: scRNA-seq defines KCs based on gene expression profiles, whereas flow cytometry differentiates between KCs and MDMs using the single surface marker, CX3CR1. Analysis of the macrophage subsets identified by scRNA-seq reveals that, while MDM clusters generally show high CX3CR1 expression, there exists a subset within MDMs, labeled *Mrc1*^{hi}, that also exhibits high levels of CX3CR1 (Supplemental Figure 9C). Consequently, if flow cytometry solely employs CX3CR1 for differentiating between KCs and MDMs, it could result in disparities when compared to scRNA-seq outcomes. Both KCs and MDMs has significantly increased in *Prdm1*^{ΔNcr1} mice, which was consist with the scRNA-seq data (Supplemental Figure 9, B and F). Despite the decrease in the proportion of Ly6c2^{hi} MDMs in *Prdm1*^{ΔNcr1} mice, the expression levels of *Ly6c2* exhibited minimal variation between WT and *Prdm1*^{ΔNcr1} mice (Supplemental Figure 9D). Intriguingly, within certain cellular subsets, notably the *Ear2*^{hi} cluster, the *Ly6c2* expression levels in KO mice were found to be higher than those in WT mice. Additionally, we employed flow cytometry to examine Ly6C expression within the macrophages. Similar with the scRNA-seq findings, there were no notable differences in Ly6C expression levels between WT and KO mice (Figure 6E; Supplemental Figure 9G).

High-resolution interactions among liver cNK cells, ILC1s, and macrophages were established and compared between *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice using the CellChat program (56). Interactions between ILC1s and total macrophages were higher than that between cNK cells and macrophages (Supplemental Figure 10, A, C, E, and G). Cross-talk between liver group 1 ILCs with macrophages enriched in macrophage migration inhibitory factor (MIF), MHC-I, CXC chemokine ligand (CXCL), Thy-1 cell surface antigen (THY1), and C-type lectin (CLEC) pathways (Supplemental Figure 10, B, D, F, and H). Although the quantity of macrophages significantly increases in *Prdm1*^{ΔNcr1} mice, there is a significant decrease in the interaction number and interaction strength between liver group 1 ILCs and macrophages (~1.5 fold) in *Prdm1*^{ΔNcr1} mice (Figure 6, F and G). The

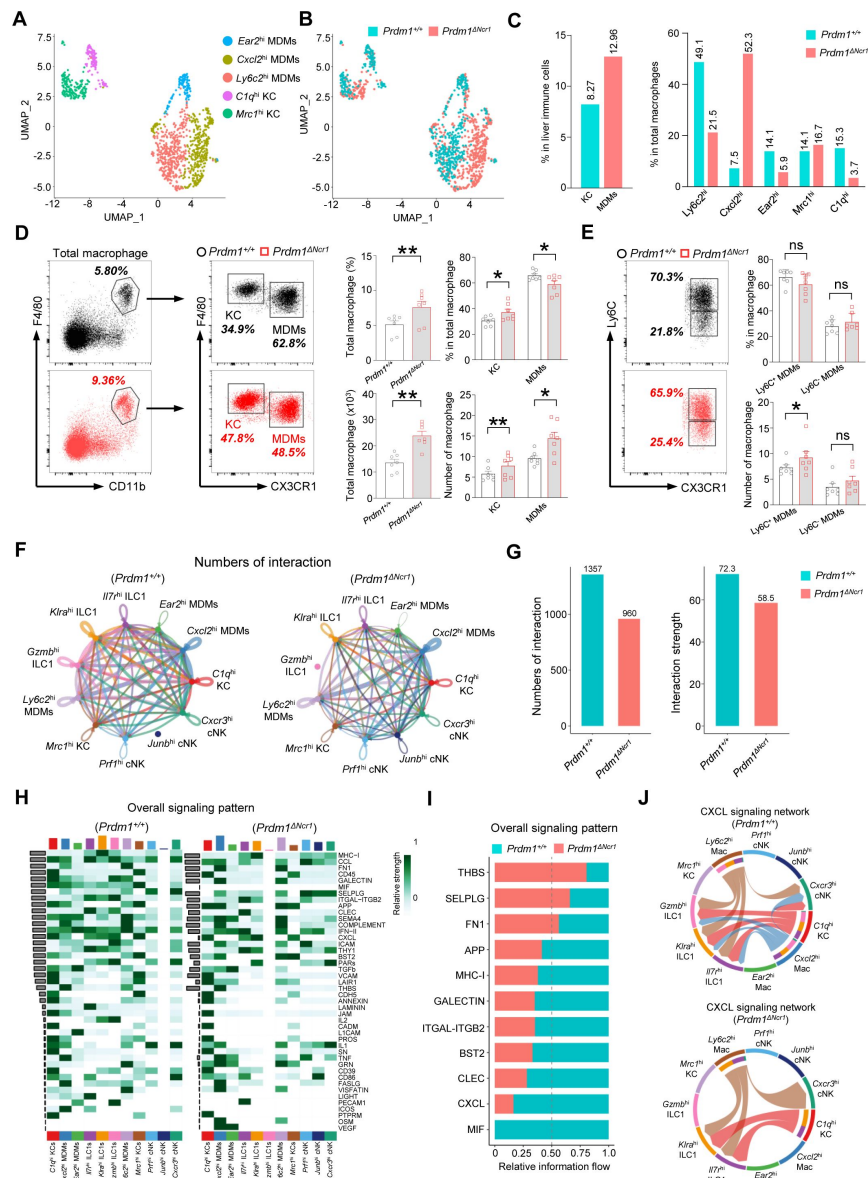


Figure 6.

Prdm1 facilitates the intercellular communication between liver group 1 ILCs and macrophages. (A and B) UMAP visualization of monocyte-derived macrophages (MDMs) and Kupffer cells (KCs) cluster (A) between *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} (B). (C) Proportions of total macrophages in liver immune cells (left), and proportions of MDMs and KCs among total macrophages in different genotypes (510 cells in *Prdm1*^{+/+}, and 624 cells in *Prdm1*^{ΔNcr1}). (D) Representative flow cytometric plots (left) and cumulative data (right) of the percentage and absolute numbers of liver total macrophages (CD45⁺Ly6G⁺CD11b⁺F4/80⁺), MDMs (CX3CR1⁺), and KCs (CX3CR1⁺) between *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} (n=7). (E) Representative flow cytometric plots (left) and cumulative data (right) of the percentage and absolute numbers of Ly6C⁺ and Ly6C⁻ cells in MDMs. (F and G) Circle plots (F) and summary data (G) illustrating the interaction numbers and strength of significant enriched ligand-receptor pairs among cluster of liver cNK cells, ILC1s, and macrophages from *Prdm1*^{+/+} (left) and *Prdm1*^{ΔNcr1} (right) cells. The thickness of the line indicates the number of enrich pairs, and the arrow reflects the direction of the interaction. (H) Heatmap of overall signaling pattern recognized from ligand-receptor pairs, which contained the sum of signaling from the sender and target cells. (I) Bar graphs showing the information flow in selected active signaling patterns between *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} cells. Relative information flow was calculated as the sum of the communication probability in given signaling patterns. (J) Chord plot of the CXCL signaling interaction network among cluster of liver cNK cells, ILC1s, and macrophages between *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} cells. Data are presented as the mean±SEM and were analyzed by 2-tailed, paired t-test. Differences were evaluated between littermates. Each circle and square on graphs represents an individual mouse; P, P-value; *, P<0.05; **, P<0.01, ns, not significant.

reduction of interaction mostly occurred in the cross-talk of ILC1-MDM and ILC1-KC, whereas no difference was observed in cNK-MDM and cNK-KC interaction (Supplemental Figure 10, A-H). A reduction in the interaction of ligand-receptor, such as Mif-CD74, Cxcl16-Cxcr6, and Cxcl10-Cxcr3 was observed in *Prdm1*^{ΔNcr1} mice compared to *Prdm1*^{+/+} mice (Supplemental Figure 11). Compared to *Prdm1*^{+/+} mice, the information flow of CXCL and MIF pathways significantly decreased in *Prdm1*^{ΔNcr1} mice (Figure 6, H and I; Supplemental Figure 10, B, D, F, and H). These pathways play a crucial role in facilitating macrophage migration. The CXCL signaling was sent from *Ly6c2*^{hi}*Cxcl2*^{hi} MDMs and *C1q*^{hi} KC, targeting all ILC1 clusters and *Cxcr3*^{hi} cNK cell clusters (Figure 6J). Of note, although the population of *Cxcl2*^{hi} macrophage primarily comprised cells from *Prdm1*^{ΔNcr1} mice, the interaction within the CXCL pathway between macrophages and group 1 ILCs was obviously less than *Prdm1*^{+/+} sample (Figure 6J). These changes could be linked to a decreased population of ILC1s and *Cxcr3*^{hi} cNK cell cluster in *Prdm1*^{ΔNcr1} mice, implying that the homeostasis of *Cxcl2*^{hi} macrophages required sufficient signals from cNK cells and ILC1s. The impaired CXCL- CXCR interactions might subsequently lead to reduced recruitment and activation of group 1 ILCs and macrophages within the tumor microenvironment.

***Prdm1* safeguards group 1 ILCs from exhaustion-like phenotypes in the tumor microenvironment**

The suppression of mitochondrial related pathways in *Junb*^{hi} cNK cell cluster, along with a significant increase of this cNK cell cluster in *Prdm1*^{ΔNcr1} mice, encouraged us to explore mitochondrial function through flow cytometry. MitoTracker, MitoSOX, and Tetramethylrhodamine methyl ester (TMRM) were used to assess the mitochondrial mass, superoxide production, and mitochondrial membrane potential. A substantial decrease in MFIs of MitoTracker was observed in *Prdm1*^{ΔNcr1} splenic cNK cells, liver cNK cells, and liver ILC1s when compared to their *Prdm1*^{+/+} counterparts (Figure 7A). This observation aligns with the enrichment of downregulated genes from *Prdm1* deficient sample in mitochondrial related pathway, as revealed by RNA sequencing data (Supplemental Figure 6, D-F). There was no significant difference in MitoSOX and TMRM between *Prdm1*^{ΔNcr1} and *Prdm1*^{+/+} mice (Figure 7B and C), which suggested that the ATP synthesize capacity was minimally affected by *Prdm1*.

IFN-γ is a critical cytokine for NK cells mediated cancer surveillance (57, 58) and impaired production of IFN-γ was considered as a key hallmark of exhausted NK cells (59, 60). To evaluate the IFN-γ secreting capacity of liver cNK cells and ILC1s in tumor microenvironment, B16F10 tumor cells were inoculated to the liver via splenic injection and the IFN-γ levels in response to stimulation of IL-12 and/or IL-18 were assessed by flow cytometry (Supplemental Figure 12A). The proportion changes of cNK cells and ILC1s in *Prdm1*^{ΔNcr1} mice was similar with the no tumor- burden condition, while the number of both cNK cells and ILC1s have significant decreased in tumor-bearing liver (Figure 7D). Compared with *Prdm1*^{+/+} mice, significant decreased of IFN-γ were observed in *Prdm1*^{ΔNcr1} mice liver cNK cells and ILC1s under the combine stimulation of IL-12/IL-18 (Figure 7E; Supplemental Figure 12B), which was more remarkable in liver ILC1s. Similar trends were observed when IL-12 or IL-18 was used alone, although only liver ILC1s showed a significant decrease in response to IL-18 stimulation (Figure 7E). These findings were consistent with the heavy tumor burden observed in *Prdm1*^{ΔNcr1} mice.

Discussion

Prdm1 is a pivotal transcription factor that has attracted substantial research interest due to its role in lymphocytes. In a study involving systemic knockout combined with competitive transplantation, it was found that *Prdm1* promotes NK cell maturation and the expression of *Gzmb*. On the contrary, the same study also found that NK cells with *Prdm1* deficiency exhibit heightened proliferation, increased survival, enhanced migratory abilities towards tumors, and greater cytotoxicity against subcutaneously implanted RMAS tumors (30). Using *Ncr1*-driven

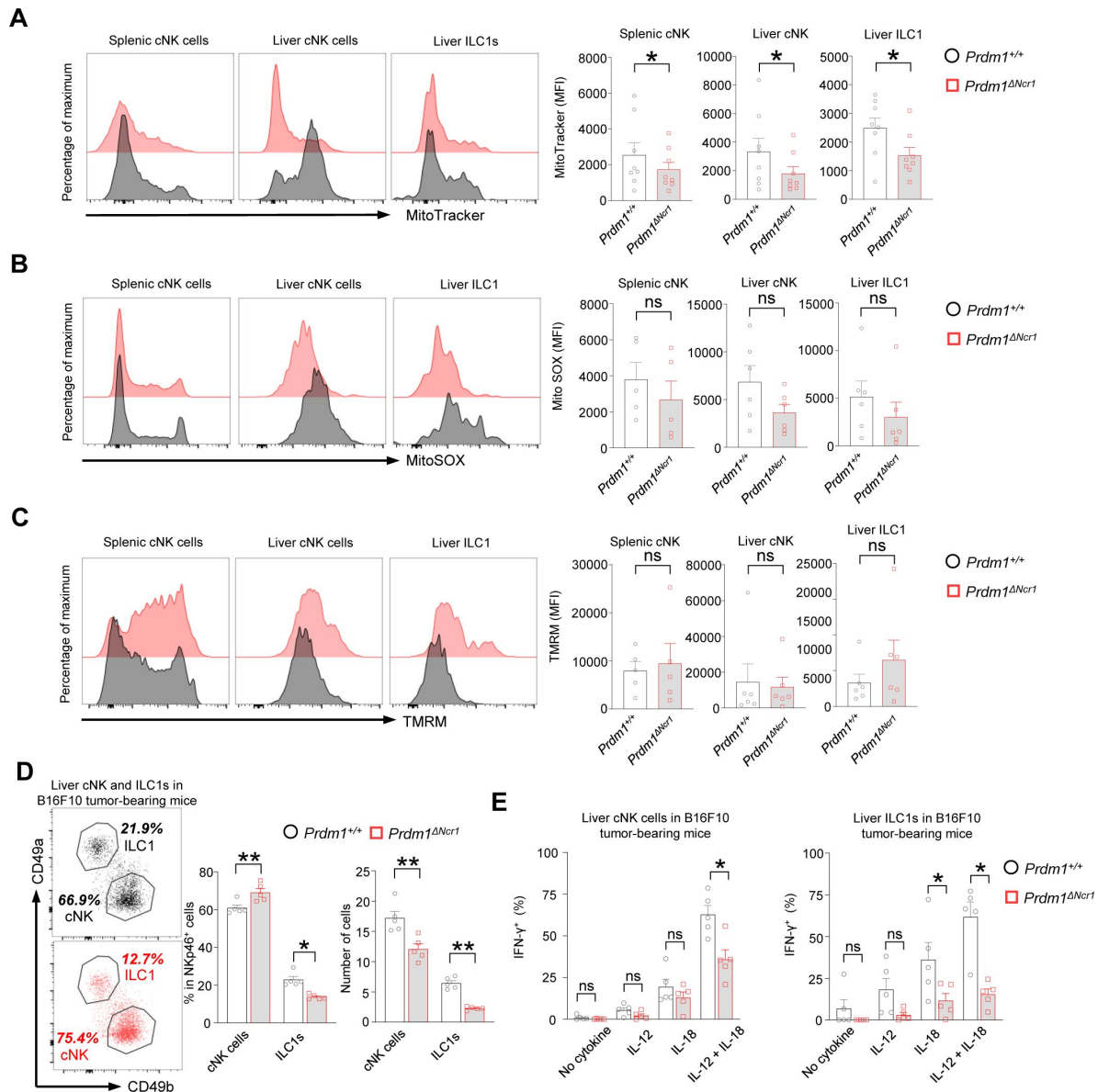


Figure 7.

Prdm1 safeguards group 1 ILCs from exhaustion-like phenotypes in the tumor microenvironment. **(A–C)** The Mitochondrial mass (MitoTracker Green staining; $n=8$) **(A)**, Mitochondrial ROS (MitoSOX staining; $n=5$) **(B)**, and Mitochondrial membrane potential (TMRM staining; $n=5$) **(C)** of splenic cNK cells, liver cNK cells and ILC1s were analyzed by flow cytometry. Representative flow cytometric plots (left) and cumulative data (right) showing the relative mean fluorescence intensities (MFIs) of each group. **(D)** Representative flow cytometric plots (left) and cumulative data (right) showing the percentage and absolute number of liver cNK cells and ILC1s in $Prdm1^{+/+}$ and $Prdm1^{\Delta Ncr1}$ tumor-bearing mice at day 14 after inoculation with B16F10 melanoma cells via intrasplenic injection ($n=5$). **(E)** Percentages of IFN- γ ⁺ liver cNK cells and ILC1s from $Prdm1^{+/+}$ and $Prdm1^{\Delta Ncr1}$ tumor-bearing mice ($n=5$). Data are presented as the mean $\pm$ SEM and were analyzed by 2-tailed, paired t-test. Differences were evaluated between littermates. Each circle and square on graphs represents an individual mouse; P , P -value; *, $P<0.05$; **, $P<0.01$; ns, not significant.

conditional knockout transgenic mice, which specifically delete *Prdm1* in group 1 ILCs, we not only validated *Prdm1*'s positive regulation of NK cell maturation, but also demonstrated its indispensable role in NK cell anti-tumor activity. The compromised mitochondrial function and reduced IFN- γ , granzyme B, and perforin production appear to be potential contributing factors to *Prdm1*-mediated cancer surveillance. Reduction of CX3CR1⁺ NK cells in multiple tissues, and decreased expression of *Cx3cr1* and *Cxcr6* was observed in *Prdm1* ^{Δ Ncr1} splenic cNK cells (**Figure 3F**), both of which are essential for NK cells egressing from bone marrow (61, 62). Our results not only confirmed a decrease in cytotoxic molecules in *Prdm1*-deficient NK cells (30) but also showed that the reduction in Gzmb and perforin is not solely attributable to the diminished maturation of these cells. Mature NK cells in bone marrow obtained the expression of CX3CR1 and acquired ability to enter the circulation (61, 62). The quantity of cNK cells increased exclusively in the bone marrow, with reductions observed in all other tissues (**Figure 1G**), indicating *Prdm1* might regulate chemokine receptors to facilitate the egression of cNK cells from the bone marrow to peripheral tissues. In addition, higher expression of *Cxcr6* compared to cNK cells is also a key factor for liver tissue residence of ILC1s (63). Decreased expression of *Cxcr6* in *Prdm1* deficient group 1 ILCs may also contribute to the balance shift towards cNK cells. Furthermore, although both liver NK cells and liver ILC1s require *Prdm1* to maintain their quantity, liver ILC1s demonstrate a more pronounced dependency on *Prdm1*. However, it is currently widely believed that liver NK cells and liver ILC1s originate from different progenitors. It is worth noting that while we observed changes in the NK and ILC1 proportions after *Prdm1* knockout, our data does not support the hypothesis that *Prdm1* affects progenitor differentiation decisions, thereby influencing the fate selection of NK and ILC1. Further research may be needed to elucidate how *Prdm1* regulates the balance between NK cells and ILC1s.

scRNA sequencing analysis reveals that both liver cNK cells and ILC1s can be further divided into three subgroups based on their gene expression patterns. *Junb* is a crucial transcriptional factor for the cytotoxic function of CD8⁺ T cells and NK cells. However, excessive *Junb* expression has been found to promote T cell exhaustion (64). Our previous study showed that as NK cells mature, the expression level of *Prdm1* increased while the expression level of *Junb* gradually decreased (25). The current study demonstrated that in NK cells, the expression level of *Junb* significantly increases upon the deletion of *Prdm1*, indicating that *Junb* expression is suppressed by *Prdm1*. As *Junb* expression decreases with NK cell maturation, and it is inhibited by the gradually increasing *Prdm1* during maturation. This implies that constraining *Junb* expression is likely a fundamental prerequisite for NK cell maturation. However, the precise mechanism by which *Prdm1* downregulates *Junb* in NK cells still needs further research. Furthermore, *Junb*^{hi} NK cells exhibit lower expression levels of cytotoxic genes and reduced mitochondrial-related signaling pathways. Mitochondria are pivotal organelles crucial for cellular metabolism. Disruptions in mitochondrial function have been linked to T Cell exhaustion, attributed to glycolytic reprogramming (65). Similarly, mitochondrial fragmentation has been closely associated with NK cell exhaustion (66). However, the concept of NK cell exhaustion isn't as firmly established as it is for T cells. Exhausted NK cells should primarily exhibit diminished functions. This is characterized by a diminished ability to destroy tumor cells, a reduced capability to activate other components of the immune system, and compromised proliferation and survival rates. Additionally, this reduced functionality is associated with a decline in the expression of molecules responsible for cytotoxic activity, lower production of IFN- γ , and metabolic disturbances that may arise from mitochondrial dysfunction. While our current data is not sufficient to definitively classify these cells as exhausted NK cells, it supports that a subpopulation, referred to *Junb*^{hi} cluster, demonstrates an exhaustion-like phenotype. The significant increase in this cell population following *Prdm1* knockout in NK cells may potentially be one of the reasons why *Prdm1* ^{Δ Ncr1} mice lose their tumor-killing capacity. Whether the excessive expression of JunB in NK cells is also a contributing factor to their exhaustion, similar to T cells (64), requires further investigation.

The scRNA-seq data revealed that *Prdm1* plays distinct roles in regulating cNK cells and ILC1s despite being required for the quantity of both lineages. Specifically, *Prdm1* appears to be more involved in promoting the resistance against exhaustion in cNK cells, whereas in ILC1s, it may play a role in the plasticity between ILC1s and ILC3s. In both our previous study and a study by Colonna et al. (25, 67), it was demonstrated that *Smad4*, a transcriptional factor involved in TGF- β signal pathway, upregulated *Prdm1* in NK cells and depletion of *Smad4* resulted in a decreased ratio of NK cells to ILC1s in the liver. However, knocking out *Prdm1* in *Ncr1*⁺ cells increased the ratio of NK cells to ILC1s in liver group 1 ILCs (Figure 2A). These findings suggest the possibility of a *Smad4*-independent pathway through which *Prdm1* promotes the maintenance of ILC1s, or that *Prdm1* plays a more significant role in maintaining ILC1s compared to its role in NK cells.

Previous studies have identified Hobit and *Prdm1* as central regulators instructing tissue-dependent programs and retention of diverse tissue-resident lymphocytes (17, 50, 52). Liver ILC1s required Hobit, but not necessary for cNK cells (6). Expression of *Prdm1* was remarkably higher in cNK cells versus ILC1s (17). While in our study, cNK cells and liver ILC1s reduced simultaneously in *Prdm1* ^{Δ Ncr1} mice, and even more significant in ILC1s. This indicates that while *Prdm1* is expressed at lower levels in ILC1s, its role in preserving the quantity of ILC1s may be more crucial. Thus, *Prdm1* and Hobit may have parallel program in instructing ILC1s functional development and maturation. *Prdm1* and Hobit directly bound and repressed *Tcf7* (17), which encoded TCF-1, a TF binding and limiting the activity of *Gzmb* regulatory element (68). *Gzmb* has been demonstrated directly bound and activated by Junb in NK cells, which suggested *Gzmb* expression regulated by multiple *Prdm1*/Hobit downstream signals (25). In human T cells, binding motif of JUNB was enriched in the binding sites of PRDM1 (69), indicating the essential role of PRDM1-JUNB axis during NK cell and T cell development. In NK cells deficient in *Prdm1* expression, we noted a decrease in *Gzmb* levels alongside with an elevation in Junb expression. This indicates that *Prdm1* not only facilitates the expression of *Gzmb* in NK cells but also suppresses Junb expression. Given that Junb is recognized as a positive regulator of *Gzmb* (70), this presents a complex interplay that seems contradictory. Therefore, it is imperative to develop a theoretical framework to comprehensively understand and interpret this paradoxical relationship. Our previous research has demonstrated that with the progression of NK cell maturation, there is a gradual increase in *Prdm1* expression, accompanied by a subtle reduction in JunB levels (25). Overexpression of JunB can induce lymphocyte exhaustion (64). Based on these findings, we hypothesize that during the initial stages of NK cell development, JunB might enhance the expression of certain molecules associated with cytotoxicity, thereby aiding NK cells in acquiring the capacity to eliminate target cells. During these initial stages, *Prdm1* level are comparatively low, thus exerting a weaker inhibitory effect on JunB. As NK cells proceed to maturity, there's a progressive increase in *Prdm1* level, which then exerts a stronger inhibitory influence on JunB, contributing to the reduced JunB level in terminal matured NK cells. Consequently, this reduces the potential for NK cell exhaustion caused by elevated levels of JunB.

Chronic inflammation is a crucial factor in promoting tumorigenesis, and macrophages play a significant role in this process. Macrophages interact with both cNK cells and ILC1s. However, the TFs that regulate these interactions are poorly understood. Fortunately, recent advances in scRNA-seq technology and the CellChat software tool (56) have allowed us to gain a better understanding of the *Prdm1* signaling pathway in group 1 ILCs and its impact on macrophages at the transcriptional level. Increased metastasis in *Prdm1* ^{Δ Ncr1} mice may be due to a decrease in the killing ability of NK cells, making them more prone to exhaustion, or it could be due to abnormalities in group 1 ILCs, leading to a decrease in the anti-tumor abilities of macrophages and an enhancement of their pro- tumor capabilities. It is worth noting that normal ILC1-macrophage interactions are more prevalent than the interaction between NK cells and macrophages (Supplemental Figure 10). We also found that CXCL signaling based interaction remarkably diminished in *Prdm1* ^{Δ Ncr1} mice, which suggested CXCL-CXCR may contribute to keep the sufficient interaction between group 1 ILC and macrophage. Specifically, *Prdm1* in group 1 ILCs may be

critical in preventing the overactivation of macrophages that can lead to cancer development. Increased population of both MDMs and KCs in *Prdm1*^{ΔNcr1} mice, as well as different distribution of macrophage clusters, indicating the homeostasis of macrophages require environment with functional cNK cells and ILC1s. Although not all macrophage phenotypes have been verified in this study, the present research serves primarily to offer initial insights and preliminary data for investigating the interactions between group 1 ILCs and macrophages. It aims to inspire further research into the role of transcription factors within the liver and the cancer microenvironment.

While our findings underscore the importance of *Prdm1* in liver cNK cells and ILC1s tumor immune surveillance, it does not be validated in human NK cells, whereas previous studies have found that PRDM1 might inhibit the proliferation and function of human NK cells (32, 71). Furthermore, we not provided an in-depth evaluation in multiple tumor models. Further research may provide deeper insight into the role of PRDM1 in the anti-tumor function of human NK cells, enabling a more direct investigation of its application in cancer therapies. Given its important role in preserving liver cNK cells and ILC1s functional heterogeneity, enhancing *Prdm1* function in human NK cells could potentially be a strategy to promote NK cell-based immunotherapy for cancer.

Methods

Mice

Prdm1^{fl/fl} mice were purchased from The Jackson Laboratory. *Ncr1-iCre* and *B2m*^{-/-} mice were purchased from Shanghai Model Organisms Center, Inc.. Six- to twelve-week-old littermates were used for the experiment.

Experimental metastasis model

For lung metastasis model, 0.3×10^6 B16F10 cells were intravenous injected into mice. Three weeks later, mice were euthanized for analysis. For liver metastasis model, mice were inoculation with 0.5×10^6 B16F10 via intrasplenic injection. Three weeks later, mice were euthanized for analysis. Lung and liver from tumor-bearing mice were fixed in 10% formalin and embedded in paraffin. Sections were stained with H&E.

In vivo cytotoxicity assay

Donor splenocytes harvested from B2m deficient (*B2m*^{-/-}) mice were labeled with 5 μM CFDA- SE. Donor splenocytes harvested from B2m-adequate (*B2m*^{+/+}) mice were labeled with 5 μM eF670. Labeled splenocytes from two mouse strains were mixed in a 1:1 ratio, and 1×10^7 cells in total were injected i.v. into *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice. One day after administration, spleen and liver cells were isolated from recipient mice, and the population of labeled cells was analyzed by flow cytometry. Rejection % was quantified according to the following formula:

$$1 - (\%B2m^{+/+} / \%B2m^{-/-})_{Prdm1^{+/+} \text{ or } Prdm1^{\Delta Ncr1}} \times 100\%.$$

Cell isolation

Mice were perfused with 10 PBS by portal vein puncture before harvesting tissues. Liver and lung was digested with 0.05% collagenase II for 30 minutes and filtered through 70 μm cell strainers, and mononuclear cells were isolated after subjected to density gradient using 30% and 70% percoll.

Spleen were also removed and pressed through 70 μm filterers to obtain splenocytes. Peripheral blood mononuclear cells were obtained from peripheral blood after lysis of red blood cells (Biolegend, 420301). Flushing femurs and mechanical disruption of inguinal lymph nodes were performed to obtain cells from bone marrow and lymph nodes.

Real-time RT-PCR

RNA was extracted from FACS-sorted NK cells or splenocytes using RNASimple Total RNA Kit (TIANGEN Biotech, 4992858) and subsequently reverse transcribed to cDNA with SuperScript VILO Master Mix (Thermo Fisher Scientific, 11755050) according to manufacturer's instructions. qPCR was performed with SYBR Green Mix (Thermo Fisher Scientific, A25742) and CFX Opus 96 Real-Time PCR System (Bio-Rad). The relative mRNA expression level was calculated using $2^{-\Delta\Delta\text{Ct}}$ method. Primer sequences: *Prdm1*: 5'-CAGAAACACTACTTGGTACA-3'; 5'-GATTGCTTGCTGCTAA-3'.

Flow cytometry

Flow cytometry and cell sorting were performed with a Cytoflex S/SRT (Beckman Coulter). The following antibodies were used (all purchased from BioLegend unless otherwise indicated): CD45-PE-Cy7 (catalog 103114, clone 30-F11); CD3 ϵ -PerCP-Cy5.5 (catalog 100327, clone 145- 2C11); NK1.1-BV421 (catalog 108741, clone PK136); CD335-AF647 (catalog 560755, clone 29A1.4, BD Bioscience); CD49a-PE (catalog 562115, clone Ha31/8, BD Bioscience); CD49b-FITC (catalog 108906, clone DX5); CD27-BV510 (catalog 124229, clone LG.3A10); CD11b-AF700 (catalog 101222, clone M1/70); KLRG1-APC (catalog 561620, clone 2F1, BD Bioscience); CD49a-BV421 (catalog 740046, clone Ha31/8, BD Bioscience); IFN- γ -PE (catalog 505807, clone XMG1.2); Granzyme B (catalog 372207, clone QA16A02); Perforin (catalog 154305, clone S16009A); CD49b-APC-Cy7 (catalog 108919, clone DX5); IgG-PE (catalog 402203, clone 27-35); CD335-BV510 (catalog 137623, clone 29A1.4); CD3 ϵ -APC-Cy7 (catalog 100330, clone 145- 2C11); CD11b-BV421 (catalog 101235, clone M1/70); CD3 ϵ -BV421 (catalog 100335, clone 145- 2C11); CD3 ϵ -FITC (catalog 553061, clone 145-2C11, BD Bioscience); CX3CR1-PE (catalog 149005, clone SA011F11); Ly6G-PerCP-Cy5.5 (catalog 127615, clone 1A8); Ly6C-BV510 (catalog 108437, clone RB6-8C5); F4/80-APC-Cy7 (catalog 123118, clone BM8). For mitochondrial metabolic assay, fresh cells were incubated in 37°C media for 30min with 100 nM MitoTracker Green (catalog M7514, Invitrogen), 100 nM TMRM (catalog T668; Invitrogen), and 10 μM MitoSOX Red (catalog M36008; Invitrogen), respectively. Surface-stained after washing with PBS and then detected by flow cytometry. For intracellular IFN- γ staining, Cells freshly obtained from liver and spleen were stimulated 12 hours with or without cytokine. GolgiStop (BD Biosciences) was added 4 hours before intracellular staining.

Bulk RNA sequencing

Total RNA from FACS sorted splenic NK cells of *Prdm1*^{+/+} and *Prdm1* ^{ΔNcr1} mice was extracted by TRIzol reagent (Invitrogen), then reverse transcribed into cDNA. Library construction was prepared using Illumina mRNA Library kit, and sequencing was performed by the BGISEQ-500. Standard methods were used to analyze the RNA-seq data, including aligning the reads to the genome by HISAT2 (v2.1.0) (72 [↗](#)), and gene expression values (Counts) were calculated using RSEM (v1.3.1) (73 [↗](#)). DEGs were identified using DEseq2 (v1.4.5) (74 [↗](#)) with a cutoff of $\log_2(\text{fold change}) > 0.5$ and $P < 0.05$. The “clusterProfiler” package (v4.4.4) (75 [↗](#)) and gene sets from molecular signatures database (MSigDB) were used for GSEA and GO analysis. The heatmap was plotted using the “Pheatmap” package (v1.0.12).

Single-cell RNA sequencing

FACS-sorted liver CD45⁺ cells with more than 80% cell viability were used for library preparation. Each sample contained cells from three *Prdm1*^{+/+} or *Prdm1* ^{ΔNcr1} mice. Gel Bead-in- Emulsions (GEMs) were generated using the 10X Genomics Chromium system, which combines Master Mix, Single Cell 3' v3.1 Gel Beads, and Partitioning Oil with single cells. GEMs were mixed with cell lysate and reverse transcription reagent to produce full-length cDNA. After incubation, the GEMs

were broken, and recovered cDNA were amplified via PCR. Fragmentation, End repair, A- tailing, and Adaptor Ligation were performed to obtain final libraries, which contain P5 and P7 sequences. The 3' library was sequenced on Novaseq 6000 with approximately 50k read pairs/cell sequencing depth. The “Seurat” R package (v4.2.0) (76 [↗](#)) was used for data analysis. Initial quality control was performed to filter out the low-quality cells and cell doublets. Cells with 200-5500 expressed genes and no more than 10% mitochondrial genes were considered for high-quality. Doublets were filtered with the “scDblFinder” Package (v1.10.0) (77 [↗](#)). After quality control, we totally recovered 6161 cells and 4817 cells from *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice, respectively. Principal component analysis (PCA) was used for cluster analysis. The first 15 PCs were picked for clustering and further visualized by UMAP. Clusters-specific marker was defined using the “FindAllMarkers” function, and clusters were manually annotated based on the top 30 or 15 markers. The “clusterProfiler” package and gene set from MSigDB were used for GSEA and GO analysis. “CellChat” package (v1.4.0) (56 [↗](#)) was utilized to predict the cell-to-cell communication from scRNA-seq data.

TCGA datasets assay

The normalized gene expression and survival datasets of cancer patients collected in The Cancer Genome Atlas (TCGA) were downloaded from UCSC Xena (<http://xena.ucsc.edu/>) (78 [↗](#)). NK cell-associated genes including *CD160*, *CD244*, *CTSW*, *FASLG*, *GZMA*, *GZMB*, *GZMH*, *IL18RAP*, *IL2RB*, *KIR2DL4*, *KLRB1*, *KLRC3*, *KLRD1*, *KLRF1*, *KLRK1*, *NCR1*, *NKG7*, *PRF1*, *XCL1*, *XCL2*, according to the previous study (33 [↗](#)). NK cell-associated genes, together with *PRDM1*, constitute the *NK-PRDM1* signature in this study. The mean expression of per genes was ordered from high-to-low and plotted by heatmap using the “Pheatmap” package. The overall survival of patients in the high and low expression of *NK-PRDM1* signature was selected for analysis. Kaplan- Meier curves were plotted by GraphPad Prism.

Statistics

For experiment results, two-tailed t tests were used to measure the continuous and normally distributed between the two independent groups. Paired t-tests were used to determine the statistical significance between two paired groups. Log-rank tests were used to compare the overall survival distribution between the two groups of patients. A P value less than 0.05 was considered significant and data were presented as mean ± SEM.

Study approval

All animal experiments were approved by The Tianjin University Animal Care and Use Committee. No human subjects were performed in this study.

Author contributions

Jitian He, and Youwei Wang designed experiments, performed experiments, analyzed the data and wrote the manuscript. Jitian He, Le Gao, Peiying Wang, Yiran Zheng, and Yumo Zhang performed experiments. Wing Keung Chan, Jiming Wang, Huaiyong Chen, and Zhouxin Yang analyzed the data, interpreted results and reviewed the manuscript.

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Declaration of interests

The authors have declared that no conflict of interest exists.

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Reviewer #1 (Public Review):

He et al. investigate the requirement and function of Blimp1 (encoded by *Prdm1*) in murine NK cells and ILC1. Employing a conditional knockout mouse model (*Prdm1*^{flox} x *Ncr1*^{cre}), the authors describe impaired abundance and maturation of *Prdm1*-deficient NK cells and ILC1 in different tissues. Blimp1-deficient NK cells have reduced expression of cytotoxic molecules (*Gzmb*, *Prf1*) and, in some instances, *Ifng* production, and *Prdm1*^{flox} x *Ncr1*^{cre} mice show impaired tumor control in experimental metastasis models. Using single cell RNA sequencing analysis, the authors propose that *Prdm1* regulates *JunB* expression and NK cell maturation. Based on in silico analyses, the authors suggest manifold intercellular communication between NK/ILC1 and macrophages. Without following up on any of these potentially interesting suggestions, the authors conclude their study reiterating that *Prdm1* regulates IFN γ -production of tumor-infiltrating NK cells and ILC1.

Many of the reported functions of Blimp1 in NK cells have previously been identified using a mixed-chimera strategy comparing *Prdm1* WT and KO NK cells (Kallies et al., Blood 2011). Here, the authors expand on these findings using a conditional model to delete *Prdm1* in NK/ILC1 and single cell sequencing, and provide a more refined analysis of the functions of Blimp1 in these cells. Cell-chat analysis suggests close interactions for Blimp-dependent NK/ILC1 subsets with hepatic macrophages, but these suggestions are not followed up by experiments. Potentially interesting differences in the macrophage compartment of *Ncr1*-Cre x *Prdm1*-fl/fl mice are suggested by the scRNA-Seq data, but are not validated e.g. by FACS. The study falls short in providing new mechanistic insights. Nevertheless, it is an interesting confirmation of "old" suggestions in a more refined setting, and the provided single-cell mRNA-Seq data represents a potentially valuable resource for the community.

<https://doi.org/10.7554/eLife.92948.2.sa1>

Reviewer #2 (Public Review):

He and colleagues aimed to elucidate the role of the transcription factor Prdm1 in liver Type 1 ILCs (innate lymphoid cells), focusing on its regulatory mechanisms and potential implications for developing innovative immune therapy strategies against liver cancer.

Strengths:

The study effectively integrates omics analyses and cytometry to explore Prdm1's impact on the cellular composition and immune regulation within the liver, providing a comprehensive view of its biological role. Employing a conditional knockout mouse model adds specificity to their experiments, allowing for precise manipulation of the Prdm1 gene.

Weaknesses:

The study predominantly relies on limited mouse models, which may not fully represent the complexity of Type 1 ILC behavior across different cancer types. Some experimental designs, such as the limited in vitro killing assessments, and additional human data could be expanded to strengthen the findings and their interpretation.

The authors have demonstrated that Prdm1 plays a critical role in the function of NK cells and ILC1s within the liver, particularly in the context of tumor resistance. However, due to the use of specific disease models and lack of direct human data, the application of these findings to clinical settings remains speculative. While the study advances our understanding of liver ILC biology, further research is necessary to validate these effects in human systems and across more diverse cancer models.

Discussion on impact and utility:

This study contributes significantly to the field of immunology and cancer therapy by revealing potential new targets for immunotherapy of liver cancer. The methods and data provided could serve as a valuable resource for further research aimed at enhancing immune-based cancer treatments.

Additional Context for Interpretation:

Understanding the role of Prdm1 in the broader context of immune cell regulation and its interaction with other cellular components in the tumor microenvironment could be crucial. Further studies should explore the dynamic between Prdm1 expression, NK cell functionality, and tumor resistance mechanisms to fully harness the therapeutic potential of targeting this pathway in liver cancer.

<https://doi.org/10.7554/eLife.92948.2.sa0>

Author response:

The following is the authors' response to the previous reviews.

Reviewer 1 (Public Review):

He et al. investigate the requirement and function of Blimp1 (encoded by Prdm1) in murine NK cells and ILC1. Employing a conditional knockout mouse model (Prdm1^{flox} x Ncr1^{cre}), the authors describe impaired abundance and maturation of Prdm1-deficient NK cells and ILC1 in different tissues. Blimp1-deficient NK cells have reduced expression of cytotoxic molecules (Gzmb, Prf1) and, in some instances, Ifng production, and Prdm1^{flox} x Ncr1^{cre} mice show impaired tumor control in experimental metastasis

models. Using single-cell RNA sequencing analysis, the authors propose that Prdm1 regulates JunB expression and NK cell maturation. Based on in silico analyses, the authors suggest manifold intercellular communication between NK/ILC1 and macrophages. Without following up on any of these potentially interesting suggestions, the authors conclude their study reiterating that Prdm1 regulates IFN γ -production of tumor-infiltrating NK cells and ILC1. Many of the reported functions of Blimp1 in NK cells have previously been identified using a mixed-chimera strategy comparing Prdm1 WT and KO NK cells (Kallies et al., Blood 2011). Here, the authors expand on these findings using a conditional model to delete Prdm1 in NK/ILC1 and single-cell sequencing and provide a more refined analysis of the functions of Blimp1 in these cells. Cell-chat analysis suggests close interactions of Blimp-dependent NK/ILC1 subsets with hepatic macrophages, but these suggestions are not followed up by experiments. Potentially interesting differences in the macrophage compartment of Ncr1-Cre x Prdm1-fl/fl mice are suggested by the scRNA-Seq data but are not validated e.g. by FACS. The study falls short in providing new mechanistic insights. Nevertheless, it is an interesting confirmation of "old" suggestions in a more refined setting, and the provided single-cell mRNA-Seq data represents a potentially valuable resource for the community. There are some control analyses that are required to support the conclusions of the authors, and I have a few suggestions that would help to improve the manuscript.

We sincerely appreciate your careful review and insightful feedback on our manuscript. We have carefully considered your comments and present the results of new experiments conducted in response to your suggestions. Please find the detailed responses below.

Major comments

Comment 1: The authors do not control for the potential effects of Cre expression. Expression of Cre from within the Ncr1 locus (using the mouse model established by Narni-Mancinelli et al.) has significant effects on NK cells and especially ILC1s (reducing their frequency and absolute numbers and altering their functionality). The authors should characterize the Ncr1cre mice used here (developed by Shanghai Model Organism Center) in this regard and should use proper controls (Ncr1Cre+ Prdm1wt/wt as control for Ncr1Cre+ Prdm1fl/fl, instead of WT littermates) for all of their key data, e.g. those depicted in Fig 1FG, 2ADF, 7D, S2,3,4.

Response 1: This is a very insightful question that has posed a challenge for many researchers, including us, engaged in conditional knockout studies. The expression of Cre and the insertion of loxP sequences both have the potential to influence gene expression. This is because the region where loxP is inserted may contain regulatory sequences for the gene of interest. Ncr1-Cre is a frequently used transgenic mouse model in our laboratory. In our prior research, we also had concerns about the possible impact of Cre on NKp46 expression, which could lead to a decline in NK cell function. Therefore, in our previous studies focused on Smad4 expression in NK cells, we conducted similar experiments. In Figure 6 of our published paper in the Journal of Clinical Investigation (Wang et al., *J Clin Invest*, 2018), we compared NKp46-iCreTgfb β 2fl/flSmad4fl/WT with NKp46-iCreTgfb β 2fl/flSmad4fl/fl. Although the primary purpose is to establish Smad4's independence from TGF- β , it also allows for a comparison between Smad4fl/fl and Smad4fl/WT in the presence of Cre. In the critical phenotype we assessed, NKp46-iCreTgfb β 2fl/flSmad4fl/fl (compared with NKp46-iCreTgfb β 2fl/flSmad4fl/WT) exhibited the same phenotype as NKp46-iCreSmad4fl/fl (compared with NKp46WTsmad4fl/fl). This suggests that Cre's influence on NK cells may be within a reasonable and controllable range. Furthermore, in contrast to the decrease in Ncr1 expression caused by Cre, the reduction in the expression levels of genes targeted by Loxp knockout, such as Prdm1 in this study (Figure 1 E), is more significant. Therefore, with the

current techniques and research methods, we believe that the data provided in this study can support the role of Prdm1 in

NK cells.

Comment 2: Several of the phenotypic findings on NK cells have been described before by Kallies et al. in 2011 (Ref 29), although using a different genetic Prdm1-ablation model (Prdm1-GFP/GFP knockin/knockout model). This study reported impaired NK cell maturation, reduced Gzmb expression, impaired in vivo cytotoxicity against subcutaneous RMA-S cells, impaired in vitro proliferation, comparable in vitro killing, increase in BM NK cell numbers. The authors should discuss/mention this more prominently in their manuscript, and highlight where they confirm or refine these previous findings, and where they actually provide new information.

Response 2: We appreciate your valuable suggestions. The article you referred to, published in *Blood*, is indeed an excellent work. While we had cited this article, our discussion regarding its specific content was limited. Based on your advice, we have made revisions and included the following content in our discussion section (page 24; line 489-493):

“In a study involving systemic knockout combined with competitive transplantation, it was found that Prdm1 promotes NK cell maturation and the expression of Gzmb. On the contrary, the same study also found that NK cells with Prdm1 deficiency exhibit heightened proliferation, increased survival, enhanced migratory abilities towards tumors, and greater cytotoxicity against subcutaneously implanted RMA-S tumors (31).”.

Comment 3: What is the reason to refer to the enriched cluster in Blimp1-deficient NK cells as "Junbhi"? There is no follow-up for a function of Junb, and there are many other genes upregulated in these cells. Most critically, these cells seem to represent exactly the c-Kithi cells that Kallies et al. already showed and discussed in their paper. The authors should stain for Kit, and also refer to this. Also, MacKay et al. performed Blimp1-Chip-Seq (in T cells), maybe it would be interesting to check to which of the identified DEGs Blimp1 can bind.

Response 3: We appreciate the suggestion from the reviewer. We think a gene that supports the development of lymphocytes doesn't necessarily positively regulate their function. For example, JunB is essential for T cell development but can also induce T cell exhaustion (Lynn et al., *Nature*. 2019). Therefore, while Prdm1 has been shown to promote NK cell development, it cannot be assumed that it always positively regulates NK cell function, especially for anti-cancer immune surveillance. In this respect, we try to find a driving-factor of the impaired anti-tumor ability of *Prdm1ΔNcr1* NK cells. Although there are many other genes upregulated in this cluster (e.g. Kit), JunB attracts more our interest of its potential for regulating NK cells functions in cancer, whereas c-Kit is more likely a marker of NK cells maturation, which has been well-demonstrated by Kallies et al. and other studies. Our previous studies also showed that the expression of c-kit was decreased in mature NK cells, compared immature NK cells (Wang et al., *J Clin Invest*, 2018).

The lack of following experiments of Junb is because we cannot find valuable surface markers to investigate the follow-up function of _Junb_hi cNK cluster. If we use intracellular markers, it is more likely an analysis of gene expression pattern, which has been well-described in our RNA-seq data. As we describe above, our study did not aim to further investigate the role of prdm1 in NK cells maturation, as the c-Kit expression was upregulated in Prdm1-knockout NK cells and correlated with NK cell maturation, which has been validated by Kallies et al..

We also have discussed the potential DEGs that could be bound and regulated by Prdm1 in our revised manuscript (page 27-28; line 561-571):

“Prdm1 and Hobit directly bound and repressed *Tcf7* (18), which encoded TCF-1, a TF binding and limiting the activity of *Gzmb* regulatory element (69). *Gzmb* has been demonstrated directly bound and activated by *Junb* in NK cells, which suggested *Gzmb* expression regulated by multiple Prdm1/Hobit downstream signals (26). In human T cells, binding motif of JUNB was enriched in the binding sites of PRDM1 (70), indicating the essential role of PRDM1-JUNB axis during NK cell and T cell development. In NK cells deficient in Prdm1 expression, we noted a decrease in *Gzmb* levels alongside with an elevation in *Junb* expression. This indicates that Prdm1 not only facilitates the expression of *Gzmb* in NK cells but also suppresses *Junb* expression. Given that *Junb* is recognized as a positive regulator of *Gzmb* (71), this presents a complex interplay that seems contradictory. Therefore, it is imperative to develop a theoretical framework to comprehensively understand and interpret this paradoxical relationship.”

Comment 4: cNK cells are considered circulating cells, that transiently pass through the liver.

Previous studies have suggested almost identical gene expression patterns in hepatic and splenic NK cells. In functional tests, they often “perform” identically. I am therefore a bit surprised that the authors find a differential dependency of Blimp1 for the IFN γ production of splenic (no role of Blimp1) versus hepatic (Blimp1 regulating IFN γ production) NK cells (Fig S3). Do the authors have any suggestions on that? The analyses are performed by 12+4h stimulations with IL12/18, which could involve the effects of altered bystander cells (as suggested by Figure 6). Therefore, these analyses should be provided upon standard 4h stimulations with IL12/18 and also with PMA/I under BFA. Note: liver and splenic cNK cells look quite different in the chosen histograms in Figures 7 A, B, C, yet there is massive variability in these analyses - is there any systematic/technical problem?

Response 4: We appreciate the valuable suggestion from the reviewer. Studies have suggested that, at the gene expression or transcriptomic level, liver NK cells exhibit more similarity to splenic NK cells while displaying greater divergence from liver ILC1s. However, we do not think that splenic NK cells or peripheral blood NK cells (which are more abundant in circulation) are entirely indistinguishable from liver NK cells. Notably, there are substantial differences in their maturity levels, with liver NK cells being more mature. Since we are examining the protein levels, a 4-hour stimulation period may not fully capture these distinctions. Even when considering the potential impact of bystander cells, the experimental design specifically targets Prdm1 knockout within NK cells, ensuring that the study accurately elucidates the role of Prdm1 in NK cells. For each experiment, we have implemented control measures, and any variances observed in the figures may be attributed to individual variations among the animals. It is also possible that the MFI values measured by flow cytometry exhibit larger variations than a percentage.

Comment 5: Figure 4 H/I - In contrast to NK cells in Fig 4E, F, the KO and WT ILC1s seem to co-cluster largely. Authors should validate differentially expressed genes. How strong is the effect of Blimp1 in ILC1s? Or is Blimp1 a critical TF driving effector differentiation in NK cells, while it has only subtle effects in ILC1 (these may be regulated by Hobit?)? This seems an interesting finding that should at least be discussed. For these types of small differences in ILC1, FACS confirmation analyses should be performed and findings be reevaluated using Cre-expressing controls (see above).

Response 5: We appreciate the suggestion from the reviewer. As request, we analyze the DEGs in liver cNK cells and ILC1s from our scRNA-seq data (revised Supplemental Figure 8, A and

B). There only a few valuable DEGs in ILC1s compared to cNK cells. It's likely that Prdm1 have more essential effect of cNK cells transcriptional program, while it plays more important role in keep the homeostasis of ILC1s population. We have discussed these points to better inform the readers. (page 27; line 554-561):

“Previous studies have identified Hobit and Prdm1 as central regulators instructing tissue-dependent programs and retention of diverse tissue-resident lymphocytes (18, 51, 53). Liver ILC1s required Hobit, but not necessary for cNK cells (6). Expression of Prdm1 was remarkably higher in cNK cells versus ILC1s (18). While in our study, cNK cells and liver ILC1s reduced simultaneously in *Prdm1ΔNcr1* mice, and even more significant in ILC1s. This indicates that while Prdm1 is expressed at lower levels in ILC1s, its role in preserving the quantity of ILC1s may be more crucial. Thus, Prdm1 and Hobit may have parallel program in instructing ILC1s functional development and maturation.”.

We cannot find valuable surface marker to evaluate the change in ILC1s, as most of changes are intracellular markers.

Comment 6: The authors describe and discuss some of Figure 1 and 2 data as if Blimp1 would be involved in alternative NK versus ILC1 fates, but there is no evidence for this.

Response 6: There is no evidence that Prdm1 could alter the fate decision of the progenitor towards liver cNK or ILC1s. Although some studies reported the conversion between cNK cells and ILC1s in special contexts, it was widely accepted that liver cNK cells and ILC1s originated from different progenitors. While we observed changes in the proportions of liver cNK cells and ILC1 in Prdm1 KO mice, we still lack sufficient evidence to support the relative independence of NK and ILC1 development, as well as evidence to indicate that Prdm1 is exclusively responsible for NK and ILC1.

Regarding the changes in NK and ILC1 proportions after Prdm1 KO, we believe that both NK and ILC1 cells require Prdm1 to maintain their populations, with ILC1 possibly requiring it to a greater extent. This is the reason for the altered balance between NK and ILC1 cells following Prdm1 KO. We wish to clarify this point to prevent any misconceptions among readers. To address this, we have added the following content to the discussion section (page 25; line 509-516):

“Furthermore, although both liver NK cells and liver ILC1s require Prdm1 to maintain their quantity, liver ILC1s demonstrate a more pronounced dependency on Prdm1. However, it is currently widely believed that liver NK cells and liver ILC1s originate from different progenitors. It is worth noting that while we observed changes in the NK and ILC1 proportions after *Prdm1* knockout, our data does not support the hypothesis that Prdm1 affects progenitor differentiation decisions, thereby influencing the fate selection of NK and ILC1. Further research may be needed to elucidate how Prdm1 regulates the balance between NK cells and ILC1s.”.

Comment 7: There are several recent studies suggesting a role for Hobit, homologue of Blimp1, in NK cells and in ILC1, and in the control of liver metastases. The authors should discuss similar and unique functions of Hobit and Blimp1, also in the regulation of gene expression patterns, and should refer to these studies.

Response 7: We would like to express our gratitude to the reviewer for your insightful comments, which bring forth a critical perspective. In accordance with the reviewer's suggestion, we have updated our discussion to include the diverse functions guided by Hobit and Prdm1 in regulating the development and function of cNK cells and ILC1s (page 27; line 554-561):

“Previous studies have identified *Hobit* and *Prdm1* as central regulators instructing tissue-dependent programs and retention of diverse tissue-resident lymphocytes (18, 51, 53). Liver ILC1s required *Hobit*, but not necessary for cNK cells (6). Expression of *Prdm1* was remarkably higher in cNK cells versus ILC1s (18). While in our study, cNK cells and liver ILC1s reduced simultaneously in *Prdm1ΔNcr1* mice, and even more significant in ILC1s. This indicates that while *Prdm1* is expressed at lower levels in ILC1s, its role in preserving the quantity of ILC1s may be more crucial. Thus, *Prdm1* and *Hobit* may have parallel program in instructing ILC1s functional development and maturation.”.

As shown in Supplemental Figure 8, we analyzed two published scRNA-seq data performed with *_Hobit_KO* mice and integrated DEGs in cNK cells and ILC1s with our data. We observed overlaps of DEGs in *Prdm1ΔNcr1* and *_Hobit_KO* between cNK cells and ILC1s, such as *Junb*, *Tcf7*, *Gzmb*, and *Prf1* (Supplemental Figure 8), indicating the similar regulatory network of *Prdm1* and *Hobit*. These data are now described on page 19; lines 386-395:

“We also compared the gene expression patterns between *Prdm1* and *Hobit* (homologue of Blimp1) with two published scRNA-seq data (51, 53). Following the knockout of *Hobit*, the DEGs were primarily identified within ILC1s. Conversely, after the knockout of *Prdm1*, a greater number of DEGs were observed in cNK cells. This indicates that *Prdm1* likely possesses a broader range of target genes within cNK cells, whereas *Hobit* appears to have a more pronounced impact on gene expression within ILC1s (Supplemental Figure 8, C-F). There are some overlaps between the downstream transcriptional profile of *Prdm1* and *Hobit* in liver cNK cells and ILC1s (Supplemental Figure 8, G and H), such as *Junb*, *Fosb*, *Tcf7*, *Kit*, *Gzmb*, *Prf1*, and *Cxcr6* was simultaneously upregulated or downregulated in both *Prdm1ΔNcr1* and *_Hobit_KO* liver cNK cells or ILC1s, indicating the similar regulatory networks of *Prdm1* and *Hobit*.”.

Comment 8: Figure 4: The authors should discuss (and cross-validate) their liver gene expression analyses in the context of published datasets of NK and ILC1, such as the ones by Lopez et al, Friedrich et al, Ducimetiere et al and Yomogida et al.

Response 8: We thank the reviewer for raising this important point. To address this question, we have now analyzed the gene expression of liver cNK cells and ILC1 in two published data mentioned above, also in the context of *Hobit*-knock. We compared gene expression of different clusters and described in our revised manuscript (page 19; lines 386-395).

“We also compared the gene expression patterns between *Prdm1* and *Hobit* (homologue of Blimp1) with two published scRNA-seq data (51, 53). Following the knockout of *Hobit*, the DEGs were primarily identified within ILC1s. Conversely, after the knockout of *Prdm1*, a greater number of DEGs were observed in cNK cells. This indicates that *Prdm1* likely possesses a broader range of target genes within cNK cells, whereas *Hobit* appears to have a more pronounced impact on gene expression within ILC1s (Supplemental Figure 8, C-F). There are some overlaps between the downstream transcriptional profile of *Prdm1* and *Hobit* in liver cNK cells and ILC1s (Supplemental Figure 8, G and H), such as *Junb*, *Fosb*, *Tcf7*, *Kit*, *Gzmb*, *Prf1*, and *Cxcr6* was simultaneously upregulated or downregulated in both *Prdm1ΔNcr1* and *_Hobit_KO* liver cNK cells or ILC1s, indicating the similar regulatory networks of *Prdm1* and *Hobit*.”.

Recommendations For The Authors:

Comment 9: The use of a paired t-test analysis when comparing cells/groups from different mice is not correct. Instead, the authors should consider using e.g. an unpaired t-test and re-test the indicated significance (e.g. Figure 1F, Figure 2H).

Response 9: We thank the reviewer's comments. As we used littermates for the experiments and they are compared side by side, so the paired t-test analysis is acceptable. We reanalysis the significance in the results of Figure 1F, and Figure 2H using unpaired t-test. The statistics significance of Figure 1F using unpaired t-test was same as using t-test. However, in Figure 2H, the reduced IFN- γ production not reach statistics significance when used un-paired t-test (Supplemental Figure 12B). It may attribute to the variation between different littermates, but the trend is still under the scope of our conclusion. We believe that employing a paired t-test between littermates could be also meaningful. As such, we kept both statistical methodologies to ensure a thorough evaluation.

Comment 10: In several instances, it is unclear whether data are pooled or representative (and if so, of how many analyses). This information needs to be provided for all analyses.

Response 10: We apologize for the lack of details and have now provided the sufficient information in our figure legends.

For example, we delete the number in original histogram to avoid the misunderstanding of the unclear whether data are pooled or representative (e.g. original Figure7 A-C; revised Figure7 A-C). Furthermore, we added the "representative" in figure legends of all flow cytometric plots to better inform readers (e.g. original Figure2, D and F; revised Figure2, B and D).

Comment 11: In the title and abstract authors use "type 1 ILCs" for both NK cells and ILC1, and it is difficult to understand which phenotypes correspond to cNK cells versus ILC1. Most of the analyses clearly separate these two different cell types. I would appreciate a lot being more accurate in the abstract, and describing cNK and ILC1 phenotypes in a clear way.

Response 11: We are really sorry for our inaccurate descriptions. According to Spits et al., (Spits et al., *Nature Reviews Immunology*, 2013) and other related studies, we have now adopted a more appropriate nomenclature as "Conventional NK cells" correspond to "cNK cells", "Type 1 innate lymphoid cells" to "ILC1s", and "Group 1 ILC" as the collective name of cNK and ILC1s.

The definition of these cells was described in the introduction (page 4, line 52-53; line58-62):

"Group 1 ILCs consist of cNK cells and ILC1s (1, 2), with distinct developmental trajectories and effect molecules (3).", "In a state of homeostasis, liver group 1 ILCs (CD45+CD3-NK1.1+NKp46+) can be discriminated into cNK cells and ILC1s by the differential expression of CD49a and CD49b (2): cNK cells are marked by the expression of CD49b, while liver ILC1s exhibit a distinctive positivity for CD49a. Tumor Necrosis Factor Related Apoptosis Inducing Ligand (TRAIL) is also expressed on liver ILC1s, but not on cNK cells (10, 11).".

We also describe cNK and ILC1 phenotypes in our scRNA-seq data, as shown in page 13; line 259-261:

"cNK cells expressed high levels of *Itga2* (CD49b) and *Eomes*, while ILC1s had high levels expression of *Itga1* (CD49a) and *Tnfrsf10* (Supplemental Figure 5, F and G).".

Comment 12: In the abstract authors state "The present study unveiled a novel regulatory mechanism of Prdm1 in liver Type 1 ILCs, showing promising potential for developing innovative immune therapy strategies against liver cancer." - maybe authors should discuss how their findings could be used for therapeutic approaches?

Response 12: We appreciate comments from the reviewer. As there hasn't been a clear consensus on the role of *Prdm1* in NK cells prior to this, some studies have suggested that *Prdm1* can inhibit cytokine secretion by NK cells. Particularly, Kallies et al. in their 2011 article in *Blood* found that *Prdm1* might suppress NK cell anti-tumor activity. Hence, there hasn't been any immunotherapy targeting *Prdm1* in NK cells for cancer treatment. Our research demonstrates the enhancing role of *Prdm1* in NK cell anti-tumor activity, providing theoretical support for NK cell therapy targeting *Prdm1*.

We added the following content to the discussion section (page 29; line 605-609):

“Further research may provide deeper insight into the role of PRDM1 in the anti-tumor function of human NK cells, enabling a more direct investigation of its application in cancer therapies. Given its important role in preserving liver cNK cells and ILC1s functional heterogeneity, enhancing *Prdm1* function in human NK cells could potentially be a strategy to promote NK cell-based immunotherapy for cancer.”.

Comment 13: The authors should explain or interpret their data a bit more (e.g. what is the consequence of GSEA enriched in negative regulation of Il6 production? (Fig. 3D) □ do NK cells produce Il6 (Figure 3)? What's the impact of Il17 signaling in NK/ILC1 (Figure 5). Do the authors suggest JunB-driven metabolic reprogramming (Suppl. Fig 6D-F?).

Response 13: We appreciate comments from the reviewer. The question of IL-6 production in NK cell also raised by another reviewer. We have checked the GSEA results, and found no valuable genes in IL-6 production in NK cells. According to the suggestions of another reviewer (Response to Reviewer 2 Comment, Comment 14), it may be prudent to omit this figure.

IL-17 signaling indicated the plasticity of ILC1s, that may be originated from the differentiation of ILC3, we added more discussion of this part (page 17; line 341-344).

“Several ILC3 signature genes, such as *Rora*, *Tmem176a*, and *Tmem176b* (45), highly expressed in this cluster (Supplemental Figure 7D). Considering the close relationship between IL-17 mediated immunity response and ILC3 (1, 46), it is plausible that *Il7r_hi* ILC1 cluster may be attributed, at least in part, to potential plasticity between ILC1 and ILC3 subsets.”.

The decreased mitochondrial function may have more relevance to NK cell exhaustion in tumors. Our data suggest that the elevated expression of JunB in NK cells may predispose them to exhaustion. Currently, our hypothesis regarding the promotion of NK cell exhaustion by high JunB expression is based on the observed correlation between JunB expression levels and exhaustion phenotypes (at the gene expression and IFN- γ secretion levels) and the findings in reference 67 (Lynn et al., *Nature*, 2019), where JunB was found to promote T cell exhaustion. However, we have not demonstrated causation between high JunB expression and exhaustion in NK cells. We propose that in NK cells, especially mature NK cells, excessive JunB expression may make them more sensitive to exhaustion inducers. Nevertheless, further research is needed to confirm this. To clarify this, we added the following content in the discussion section (page 26; line 537-543):

“While our current data is not sufficient to definitively classify these cells as exhausted NK cells, it supports that a subpopulation, referred to *Junbhi* cluster, demonstrates an exhaustion-like phenotype.

The significant increase in this cell population following *Prdm1* knockout in NK cells may potentially be one of the reasons why *Prdm1* Δ *Ncr1* mice lose their tumor-killing capacity. Whether the excessive expression of JunB in NK cells is also a contributing factor to their exhaustion, similar to T cells(65), requires further investigation.”.

Comment 14: Ref 25 and Ref 57 are the same publication?

Response 14: We are really sorry for our careless mistakes. We have checked all the reference and corrected the wrong format.

Comment 15: Figure 1, E - The method description of RT-PCR is missing. I apologize if I have overlooked this information.

Response 15: We have now added the description of RT-PCR in our revised method section (page 31; line 638-644):

“RNA was extracted from FACS-sorted NK cells or splenocytes using RNASimple Total RNA Kit (TIANGEN Biotech, 4992858) and subsequently reverse transcribed to cDNA with SuperScript VILO Master Mix (Thermo Fisher Scientific, 11755050) according to manufacturer’s instructions. qPCR was performed with SYBR Green Mix (Thermo Fisher Scientific, A25742) and CFX Opus 96 Real-Time PCR System (Bio-Rad). The relative mRNA expression level was calculated using 2- $\Delta\Delta C_t$ method. Primer sequences: Prdm1: 5’-CAGAAACACTACTTGGTACA-3’; 5’-GATTGCTTGTGCTGCTAA-3’.”

Comment 16: Figure 1, F - The NKp46+CD3- gate for the liver seems to cut the population, not all cells are included.

Response 16: We appreciate the review’s comment and apologize for our carelessness. We expand our data with more samples and reanalyzed them with a more convincing gating strategy. We now update our figures (revised Figure 1G; revised Supplemental Figure 2A). Several changes have occurred in the data and conclusions, and we have accordingly revised these contents in our manuscript.

The original text is:

“Proportion and absolute number of cNK cells in blood, bone marrow, lung, liver, spleen, and lymph nodes were analyzed by flow cytometry. Compared with *Prdm1*^{+/+} mice, the percentage of cNK cells (CD3-NK1.1+NKp46+) among lymphocytes was decreased in all of these tissues except bone marrow and lymph nodes (Figure 1F; Supplemental Figure 2A). However, no significant difference was observed in the percentage of cNK cells among bone marrow-derived lymphocytes between *Prdm1 Δ Ncr1* and *Prdm1*^{+/+} mice. The absolute number of cNK cells in blood, lung, liver, and spleen also decreased in *Prdm1 Δ Ncr1* mice (Figure 1F; Supplemental Figure 2A). Only a slight decrease in the number of cNK cells was observed in the lymph nodes of *Prdm1 Δ Ncr1* mice, which did not reach statistical significance either (Supplemental Figure 2A). In contrast, the absolute number of cNK cells in *Prdm1 Δ Ncr1* mice bone marrow is moderately higher than *Prdm1 Δ Ncr1* mice (Figure 1F).”

The revised text is (page 8; line 142-146):

“Proportion and absolute number of cNK cells in blood, bone marrow, lung, liver, spleen, and lymph nodes were analyzed by flow cytometry. Compared with *Prdm1*^{+/+} mice, the percentage and absolute number of NK cells (CD45+CD3-NK1.1+NKp46+) among lymphocytes was decreased in all of these tissues, whereas increased number of NK cells were observed in bone marrow (Figure 1G; Supplemental Figure 2A).”

Comment 17: Figure 1, The y-axis labeling of lung CD3-NKp46+ cells ($\times 10^3$) is not correct.

Response 17: We are really sorry for our carelessness. We now check the labels and make sure they are correct.

Comment 18: Figure 1, The statistical significance of absolute numbers of NKp46+ cells in the bone marrow should be reviewed.

Response 18: We expand our data with more samples and reanalyzed them with a more convincing gating strategy. We observed significant increase of bone marrow NK cells quantity in our updated data. These changes are now described in our revised manuscript.

The original text is:

“However, no significant difference was observed in the percentage of cNK cells among bone marrow-derived lymphocytes between *Prdm1ΔNcr1* and *Prdm1*^{+/+} mice”, “In contrast, the absolute number of cNK cells in *Prdm1*^{fl/fl} mice bone marrow is moderately higher than *Prdm1ΔNcr1* mice (Figure 1F).”

The revised text is (page 8; line 142-146):

“Proportion and absolute number of cNK cells in blood, bone marrow, lung, liver, spleen, and lymph nodes were analyzed by flow cytometry. Compared with *Prdm1*^{+/+} mice, the percentage and absolute number of NK cells (CD45+CD3-NK1.1+NKp46+) among lymphocytes was decreased in all of these tissues, whereas increased number of NK cells were observed in bone marrow (Figure 1G; Supplemental Figure 2A).”

Comment 19: Figure 1, G - CD27 and CD11b are used to define maturation stages within NK cells. Here the authors are analyzing group 1 ILC instead (containing both NK cells and ILC1, especially in the liver). It would be better to pre-gate on Eomes+ or CD49b+ NK cells for this analysis.

Response 19: We apologize for the lack of details in this analysis. We have pre-gate CD49b+ NK cells for the maturation stages analysis. We have now added this statement in our revised manuscript and figure legend (page 8; line 149-151)

“The maturation of cNK cells (gated by CD45+CD3-NK1.1+NKp46+CD49b+) from blood, bone marrow, lung, liver, spleen, and lymph nodes were assessed, based on the expression of CD11b and CD27.”.

Comment 20: Supplementary Figure 1, A - The NKp46+CD3- gate seems to cut the population, not all cells are included. y-axis labeling of spleen CD3-NKp46+ cells (%) is not correct.

Response 20: Thanks, we have corrected these errors and shown in our revised supplementary Figure 2A.

Comment 21: Figure 2, D-G - Did the authors analyse the ILC1/NK compartment of the tumor? What is the abundance and phenotype of these cells dependent on Prdm1 expression? Proper Crecontrols should be used (see above).

Response 21: We appreciate the suggestions from the reviewer. As request, we have now added the analysis of cNK/ILC1s population in the context of tumor. The proportion changes of cNK cells and ILC1s in *Prdm1ΔNcr1* mice was similar with the no tumor-burden condition, while the number of both cNK cells and ILC1s decreased in tumor bearing liver (revised Figure 7D). These contents have been updated in our revised manuscript (page 23; line 479-481):

“The proportion changes of cNK cells and ILC1s in *Prdm1ΔNcr1* mice was similar with the no tumorburden condition, while the number of both cNK cells and ILC1s have significant

decreased in tumor-bearing liver (Figure 7D).”.

The reason why we did not use Cre-controls was described in comment 1.

Comment 22: Figure 2, H - Prdm1-deficient NK and ILC1 produce less Ifng in response to in vitro stimulations with IL-12 and /or IL-18, and bulk Seq analysis (Fig 3F) shows reduced Il12rb2 expression. Does the expression of cytokine receptors correlate with the maturation of NK cells? This could be analyzed from the single-cell RNA-seq dataset. The statistical significance of %Ifng after IL12/IL18 stimulation should be revisited (see above).

Response 22: We thank the reviewer for the suggestions. To address this question, we explored the expression of IL-12 and IL-18 receptors in cNK and ILC1 clusters. Within cNK clusters, *Il12rb2*, *Il18r1* and *Il18rap* was highly expressed in *Prf1hi* and *Cxcr3hi* cNK clusters (revised Supplemental Figure 6H), indicating the IL-18 receptor expression correlated with the NK cell maturation. While in ILC1, these receptors mostly expressed on *_Il7r_hi* and *_Gzmb_hi* ILC1 clusters (revised Supplemental Figure 7C). Significant decreased of *Il18r1* expression in *Prdm1ΔNcr1* cNK cells and ILC1s may associated with the impaired ability to produce IFN-γ. We now added this analysis (page 18; line 364-368):

“Within cNK cells, *Il12rb2*, *Il18r1* and *Il18rap* was highly expressed in *Prf1hi* and *Cxcr3hi* cNK clusters (Supplemental Figure 6I), indicating the IL-18 receptor expression correlated with the NK cell maturation. While in ILC1, these receptors mostly expressed on *_Il7r_hi* and *_Gzmb_hi* ILC1 clusters (Supplemental Figure 7D). Significant decreased of *Il18r1* expression in *Prdm1ΔNcr1* cNK cells and ILC1s may associated with the impaired ability to produce IFN-γ.”.

The un-paired t test of IFN-γ production was displayed in revised supplemental Figure 12 B. Difference in IFN-γ production was found to be not significant when analyzed using an unpaired ttest in original Figure 2 H. However, significance was observed in tumor-bearing liver cNK cells and ILC1s, specifically under the context of IL-12/IL-18 stimulation, as depicted in the original Figure 7E using an unpaired t-test. These variations may be attributed to differences among different littermates. Despite these variations, the trend remains consistent with our overall conclusions. We believe that employing a paired t-test between littermates could be also meaningful. As such, we kept both statistical methodologies to ensure a thorough evaluation.

Comment 23: Figure 3, A-E - For bulk sequencing analysis, splenic CD3-NK1.1+NKp46+ were isolated. This population also contains ILC1 in the spleen (e.g. Flommersfeld et al.), although much less abundant compared to NK cells, and compared to the liver compartment. However, have the authors tested the abundance of splenic ILC1 in Prdm1-deficient mice, which may impact the gene expression data? In line with this the detection of altered Cxcr6 expression in Figure F, which is usually expressed by ILC1 rather than NK cells, may indicate an alteration in ILC1 numbers. The authors should validate the altered expression of CXCR6, Itga1, and Cx3cr1 on NK cells by flow cytometry.

Response 23: We cited the work of Flommersfeld et al. into our manuscript and have expanded our Results section to include the following information (page 19; line 377-385):

“Previous research found that spleen NK cells could be divided into three distinct groups based on their expression levels of CD27, CD62L, CD49a, and CD49b (52). CD27+CD62L- NK cells have remarkable high expression of Batf3, while it was only barely expressed in CD27+CD62L+ and CD27-CD62L+ NK cells (52). Based the sequencing data published by Flommersfeld et al., (GSE180978), a notable negative correlation was observed between the expression levels of Prdm1 and Batf3 (Supplemental Figure 8I). On top of that, our findings

unveiled the negative regulatory influence of Prdm1 on Batf3 within both spleen and liver NK cells. This discovery highlights a potential upstream mechanism that may influence the hemostasis of the spleen NK cell subpopulations through Batf3.”.

We validated the expression of CD49a (Itga1) and CX3CR1 in liver cNK cells and ILC1s in our revised manuscript, which is described in our revised manuscript (page 9; line 170-174, page 14; line 231-233):

“Increased CD49a expression was also observed in *Prdm1ΔNcr1* liver ILC1s, while it showed decreased expression in NKp46+ cells in the liver, bone marrow, and lymph nodes (Supplemental Figure 2, F and G).”, “The percentage of CX3CR1+ cNK cells was significantly decreased in multiple tissues of *Prdm1ΔNcr1* mice, while the proportion of CX3CR1+ ILC1 was increased in the liver (Figure 3F).”

Comment 24: Figure 3, F - Tnfsf26: which gene is this? is this a typo? Is a function of this gene in NK cells reported? Altered Batf3 expression suggests an impact on ILC1-like NK cells (Flommersfeld et al).

Response 24: We are very sorry for our mistakes. We have removed Tnfsf26 from the heatmap.

Comment 25: Figure 3, G-J refer to Kallies data?!

Response 25: Kallies’s data has mentioned the reduced GzmB expression in Blimp1gfp/gfp mice. However, compared with Kallies’s study, we further analyzed the GzmB and Perforin expression in different mature stages of NK cells. Reduced GzmB expression not only due to the less mature phenotype in *Prdm1*-deficient NK cells, highlighting the role of Prdm1 in regulating NK cell function. So, we added these contents in the revised manuscript (page 12; line 233-242):

“Lower GZMB and PRF1 production was observed in *Prdm1*-deficient splenic cNK cells, liver cNK cells and ILC1s (Figure 3, H-K; Supplemental Figure 4, A-I). Notably, the proportion of GZMB+ and PRF1+ cNK cells was decreased among almost all of the maturation stages of cNK cells (Figure 3, J and K). The relative mean fluorescent intensities (MFIs) of GZMB and PRF1 consistently show a reduction across all developmental stages in *Prdm1ΔNcr1* NK cells (Supplemental Figure 4, H and I). Yet, no statistical difference of PRF1 was found within the CD11b-CD27+ and CD11b+CD27+ subsets, likely due to the relatively lower perforin levels in these populations (Supplemental Figure 4I). These findings suggest that Prdm1 may directly influence cytotoxic molecule in NK cells, rather than impacting their anti-tumor abilities solely by affecting the maturation phenotype of Prdm1-deficient NK cells.”

In Discussion section (Kallies’s work is cited here in revised manuscript) (page 24; line 500-502):

“Our results not only confirmed a decrease in cytotoxic molecules in *Prdm1*-deficient NK cells (31) but also showed that the reduction in Gzmb and perforin is not solely attributable to the diminished maturation of these cells.”

Comment 26: Figure 3, G, I - How do the authors explain the high variability of GzmB and Prf1 in Prdm1+/+ cells? 2 samples have comparable values to Prdm1-deficient cells.

Response 26: This may be due to the inherent differences in MFI among different samples. In the revised version, we have added data on percentages, which exhibit much less variability (Figure 3, H and I). The MFIs of GZMB and PRF1 are moved to supplemental Figure 4 E and F.

Comment 27: Did the authors test the mice for potential germline recombination of the floxed allele, which has been suggested as a potential problem of Ncr1cre?

Response 27: We appreciate the insightful comments provided by the reviewer, and this is a really good question. In *Prdm1^{fl/fl}* mice, germline recombination typically results in a systemic knockout of *Prdm1*, which can lead to embryonic lethality. Given that mice were successfully born in the current study, it is almost unlikely that germline recombination of *Prdm1* occurred due to leaky expression of Cre.

To confirm this issue, we isolated splenocytes and assessed *Prdm1* expression using qPCR. We observed no significant difference in *Prdm1* expression between splenocytes from *Prdm1^{+/+}* and *Prdm1ΔNcr1* mice (revised Figure 1F). This also indicated that germline recombination issues are unlikely to be present in the *Prdm1ΔNcr1* mice.

Comment 28: Histograms do not show MFI

Response 28: We appreciate the comments provided by the reviewer. The MFI value was omitted.

Comment 29: Supplementary Figure 4, B - FACS plot labelling: Typo, Histograms do not show MFI.

Response 29: We sincerely thank the reviewer for careful reading. The typo in this figure was corrected. The MFI is omitted.

Comment 30: Figure 4, A - What are the cells in the red cluster in the middle of the UMAP, do they belong to B cells? Why do they cluster so separately? It is interesting, but also surprising that NK and ILC1 cluster map so far apart from each other (rather with CD8 or B cells? or NKT cells) - do the authors have any comments?

Response 30: We sincerely apologize for the mistakes in labeling a group of cells in our previous analysis. Upon a thorough re-evaluation, we have corrected the labels of several cell clusters that were previously misidentified. The revised heatmap (revised Supplemental Figure 5C) represents the marker genes for each cluster. Additionally, in our updated analysis (revised Figure 4A), we have included clusters for Epithelial cells, CD4⁺ T cells, NKT cells, and Kupffer cells. Please note, the red cluster identified in the center of the original heatmap corresponds to the CD4⁺ T cells.

We checked the markers of cNK cell and ILC1 clusters and confirmed they are labeled correctly, as *Ncr1* and *Klrb1c* (NK1.1) was highly expressed in these clusters compared to others (revised Supplemental Figures 5E).

Comment 31: Does Junb expression correlate with the maturation stages of NK cells?

Response 31: Our previous research indicated that during the maturation process of NK cells, there was a decrease in the expression levels of *Junb* (negative correlation), whereas there was an increase in the expression levels of *Prdm1* (Wang et al., *J Clin Invest*, 2018; Supplemental Figure 5c and Supplemental Figure 11).

Comment 32: The authors may consider validating their scRNA-seq data (e.g. by FACS analysis for highlighted markers, eg. cKit, Tcf7, Gzma, Cxcr3).

Response 32: We appreciate the suggestion from the reviewer. We validated several marker genes, including *Gzmb*, *Prf1*, and *Cx3cr1* by FACS, as shown in the revised Figure 3 F-K.

Currently, FACS cannot distinguish liver NK cells into as many distinct clusters as can be achieved through scRNAseq analysis. However, we expect that as technology progresses, we will be able to enhance our validation of the scRNA-seq data.

Comment 33: It is a bit unclear to me why authors refer to Cxcr3hi NK cells as tissue-resident. This is based on Cxcr3 and Ccr2 expression. To make this statement, a much more detailed analysis would be required. How are CD69, CD49a, or CXCR6 expression of these cells?

Response 34: We appreciate the suggestion from the reviewer. The primary reason for classifying this specific cluster of NK cells as tissue-resident is derived from the differential expression genes (DEGs) and Gene Ontology (GO) analysis, which demonstrate significant chemokine receptor activity within this cluster.

To make this statement more clearly, we check the expression of the above markers, but only Cd69 had expression in cNK clusters, which was highly expressed in _Junb_hi and _Cxcr3_hi cNK cells (revised Supplemental Figure 6D). We also used top30 DEGs in ILC1s versus cNK to calculate the module score in all cNK clusters, as _Cxcr3_hi cNK had highest score among these clusters (revised Supplemental Figure 6D). This part has been updated in our manuscript (page 15; line 298-308):

“Expression of tissue-resident markers Cd69 was also highly expressed in this clusters (Supplemental Figure 6D). The enrichment of chemokine receptors in the genes upregulated in the _Cxcr3_hi cluster implying a greater likelihood of this cluster being tissue-resident compared with other cNK cell clusters (Figure 4H). To further confirmed tissue-resident properties of this clusters, we calculated the module score based on top30 DEGs in ILC1 versus cNK clusters, including *Cxcr6*, *Itga1*, *Cd160*, *Cd226*, etc. _Cxcr3_hi cNK clusters have the highest score among all cNK clusters (Supplemental Figure 6H), indicating the similarity with liver ILC1s. In the tumor microenvironment, reports indicated that NK cells could transform into ILC1s (25). If this conversion of cNK cells into ILC1s also occurred under normal physiological conditions, then _Cxcr3_hi cNK cell cluster might be the most susceptible to such transformation.”

Comment 35: The authors suggest that Prdm1 regulates chemokine receptor expression. An alternative explanation could be that this is an indirect effect of altering the abundance of NK cell subsets.

Response 35: We are sorry for lacking the details in these figures. The input cell number of each genotype has now been added in following figure legends.

Figure 4F, “Proportions of cNK cells among total cNK cells (left; 211 cells in *Prdm1*^{+/+}, and 141 cells in *Prdm1ΔNcr1*) and within clusters (right).”; Figure 5C, “Proportions of ILC1s among total ILC1s in different genotypes (left; 114 cells in *Prdm1*^{+/+}, and 63 cells in *Prdm1ΔNcr1*) and within each cluster (right).”; Figure 6C, “Proportions of MDMs and KCs among total macrophages in different genotypes (510 cells in *Prdm1*^{+/+}, and 624 cells in *Prdm1ΔNcr1*).”

To minimize the effects of discrepancies in input numbers between samples with different genotypes, we represented the relative proportions of each cluster within its specific genotype (e.g. Supplemental Figure 6B; Supplemental Figure 7B; Supplemental Figure 9B).

Comment 36: Supplementary Figures 6 and 7, A - The formatting of gene annotations does not fit the heat maps (the gene names on the last rows are missing).

Response 36: We apologize for our careless mistakes. We have now addressed these mistakes.

Comment 37: Supplementary Figures 6 and 7, What is the consequence of compromised mitochondrial function? Increase apoptosis?

Response 37: In our experiments, we did not find that *Prdm1* has an effect on the apoptosis of NK cells. Conversely, previous studies have found that *Prdm1* might inhibit the proliferation of NK cells (C. Kucuk, et. al., PNAS, 2011). We acknowledge that there is ongoing debate regarding the precise definition of NK cell exhaustion. In our experiments, no changes were detected in the expression levels of surface markers (TIGIT) associated with exhaustion on NK cells following the knockout of *Prdm1*. However, we did note a significant reduction in the cytokine secretion capacity and tumor control efficacy of NK cells after *Prdm1* knockout. We prefer to say that the consequence of compromised mitochondrial function might be increased exhaustion. As we mentioned in discussion part (line 482-483), mitochondrial fragmentation has been confirmed to be closely associated with NK cell exhaustion in tumor (Zheng et al. *Nature immunology*, 2019). Although the evidence to define the exhausted NK cells in *Prdm1ΔNcr1* was not sufficient, our data may support the compromised mitochondrial functions, at least in part, associated with the exhausted phenotype of *Prdm1ΔNcr1* NK cells in cancer.

We have discussed these points in our revised manuscript (page 26; line 529-543):

“Mitochondria are pivotal organelles crucial for cellular metabolism. Disruptions in mitochondrial function have been linked to T Cell exhaustion, attributed to glycolytic reprogramming (66). Similarly, mitochondrial fragmentation has been closely associated with NK cell exhaustion (67).

However, the concept of NK cell exhaustion isn't as firmly established as it is for T cells. Exhausted NK cells should primarily exhibit diminished functions. This is characterized by a diminished ability to destroy tumor cells, a reduced capability to activate other components of the immune system, and compromised proliferation and survival rates. Additionally, this reduced functionality is associated with a decline in the expression of molecules responsible for cytotoxic activity, lower production of IFN- γ , and metabolic disturbances that may arise from mitochondrial dysfunction. While our current data is not sufficient to definitively classify these cells as exhausted NK cells, it supports that a subpopulation, referred to *_Junb_hi* cluster, demonstrates an exhaustion-like phenotype. The significant increase in this cell population following *Prdm1* knockout in NK cells may potentially be one of the reasons why *Prdm1ΔNcr1* mice lose their tumor-killing capacity. Whether the excessive expression of JunB in NK cells is also a contributing factor to their exhaustion, similar to T cells(65), requires further investigation.”.

Comment 38: Figure 5, Describing the scRNA Seq data, the authors are switching a lot between Figure 4 and Figure 5. Maybe a reorganization of the Figures (Figure 4: NK cell; Figure 5: ILC1) could help.

Response 38: We appreciate the reviewer's suggestion. We have now reorganized the Figure 4 and Figure 5.

Comment 39: Figure 5, We suggest naming one of the ILC1 clusters "Gzmbhi" to keep it consistent with the FACS data.

Response 39: We agree with this excellent suggestion and have now renaming the “Gzmahi” ILC1 cluster as “Gzmbhi” ILC1 cluster.

Comment 40: Figure 5, C - How was the JunB score derived (which genes were used)?

Response 40: The JunB score was calculated based on the expression of marker genes in _Junb_hi cNK clusters (DEGs in _Junb_hi cNK cluster compared to other clusters, as shown in revised Supplemental figure 6A). The score was calculated using “AddModuleScore” R package.

Comment 41: Figure 5, G, I - The authors highlight Il17 signaling pathway, what is the impact of Il17 on NK/ILC1? Did the authors check for ILC3 (Rorc expression) within the ILC1 cluster?

Response 41: The enrichment of IL-17 signaling pathway in _Il7r_hi ILC1 indicated that this cluster encompass ILC1s originate from the conversion of Rort+ ILC3s. Although the Rorc expression was undetectable in all ILC1 clusters, we found several ILC3 marker genes highly expressed in this clusters (e.g. Rora, Tmem176a, Tmem176b) according to the ILC3 transcriptomes (Robinette et al., *Nature Immunology*, 2015).

We have added these contents in our revised manuscript (page 17; line 341-344):

“Several ILC3 signature genes, such as *Rora*, *Tmem176a*, and *Tmem176b* (45), highly expressed in this cluster (Supplemental Figure 7D). Considering the close relationship between IL-17 mediated immunity response and ILC3 (1, 46), it is plausible that _Il7r_hi ILC1 cluster may be attributed, at least in part, to potential plasticity between ILC1 and ILC3 subsets.”.

Comment 42: Figure 5, The authors detect more Ly49E+ cytotoxic ILC1 in Prdm1fl Ncr1cre mice.

How does this observation fit to the reduced cytotoxicity of NK cells?

Response 42: The proportion of _Klra_hi ILC1 was increased, while the _Gzmb_hi ILC1 was decreased in *Prdm1ΔNcr1* mice. Moreover, total number of three ILC1 cluster was reduced in *Prdm1ΔNcr1* mice.

Comment 43: Line 350/351: Citation required.

Response 43: We added the respective reference. (reference 55 and 56).

Comment 44: Figure 6, The Cell-chat analysis provides interesting suggestions, but none are experimentally addressed. It is also difficult to evaluate these analyses: are any of the Mac subsets altered in frequency or phenotype in either genotype? This could be analyzed from the single-cell data in Fig 4. At the very least, flow cytometric validation of predicted shifts in the Mac compartment should be confirmed.

Response 44: We gratefully thanks for these valuable suggestions. As requested, we analyzed macrophages and validated some of the scRNA-seq data by flow cytometry. We have re-written this part with the analysis of altered proportion of two macrophage clusters (Kupffer cells and Monocyte-derived macrophages) (page 20-21; line 399-436):

“The scRNA sequencing analysis identified two well-established subpopulations of liver macrophages: the resident Kupffer Cells (KCs) and the Monocyte-Derived Macrophages (MDMs) (Figure 6, A-C; Supplemental Figure 9A). When comparing the total proportion of macrophages within the immune cell population of the liver between WT and *Prdm1ΔNcr1* mice, there is an increase in *Prdm1ΔNcr1* mice (Figure 6C). To confirm these findings, we utilized flow cytometry to define macrophages, including both KCs and MDMs, gating by CD45+Ly6G-F4/80+CD11b+ (Figure 6D).

Our analysis showed that, following the deletion of *Prdm1* in Group 1 ILCs, there is a significant increase in both the proportion and number of macrophages in the liver (Figure 6D).

According to the transcriptional profile, liver macrophages further clustered and were labeled as “_Ly6c2_hi”; “_Cxcl2_hi”; “_Ear2_hi” MDMs, and “_Mrc1_hi”; “_C1q_hi” KCs (Figure 6A, Supplemental Figure 9, A-E). Increased proportion of MDMs and KCs was observed in *Prdm1ΔNcr1* cells (Supplemental Figure 9B). Within MDMs clusters, _Ly6c2_hi MDMs mainly compose of *Prdm1*^{+/+} cells, while *Prdm1ΔNcr1* cells concentrated in _Cxcl2_hi cluster (Figure 6C). The scRNA-seq data reveal that following *Prdm1* knockout in NKp46⁺ cells, there is a decrease in the proportion of KCs within the macrophage population, while the proportion of MDMs increases (Figure 6D). CX3CR1, a chemokine receptor, is extensively utilized to distinguish KCs and MDMs within macrophages. Cells expressing CX3CR1 are identified as MDMs, whereas those without CX3CR1 expression are categorized as KCs (56). Employing flow cytometry and leveraging CX3CR1 expression, we assessed the ratios of KCs and MDMs. However, diverging from the scRNA-seq findings, flow cytometry indicates that post-*Prdm1* knockout in group 1 ILCs, there is a minor increase in the proportion of KCs within the total liver macrophages, and a decrease in the proportion of MDMs (Figure 6D; Supplemental Figure 9B). This discrepancy could stem from the different bases of classification: scRNA-seq defines KCs based on gene expression profiles, whereas flow cytometry differentiates between KCs and MDMs using the single surface marker, CX3CR1. Analysis of the macrophage subsets identified by scRNA-seq reveals that, while MDM clusters generally show high CX3CR1 expression, there exists a subset within MDMs, labeled *Mrc1hi*, that also exhibits high levels of CX3CR1 (Supplemental Figure 9C). Consequently, if flow cytometry solely employs CX3CR1 for differentiating between KCs and MDMs, it could result in disparities when compared to scRNA-seq outcomes. Both KCs and MDMs has significantly increased in *Prdm1ΔNcr1* mice, which was consist with the scRNA-seq data (Supplemental Figure 9, B and F). Despite the decrease in the proportion of *Ly6c2hi* MDMs in *Prdm1ΔNcr1* mice, the expression levels of *Ly6c2* exhibited minimal variation between WT and *Prdm1ΔNcr1* mice (Supplemental Figure 9D). Intriguingly, within certain cellular subsets, notably the *Ear2hi* cluster, the *Ly6c2* expression levels in KO mice were found to be higher than those in WT mice. Additionally, we employed flow cytometry to examine Ly6C expression within the macrophages. Similar with the scRNA-seq findings, there were no notable differences in Ly6C expression levels between WT and KO mice (Figure 6E; Supplemental Figure 9G).”.

The changes of the macrophage compartment indicated the potential influence of functional NK cells to macrophages. We have revised these parts in our results and discussion (line 590-601). However, to address more analysis on macrophage is worthy but would go beyond the scope of this manuscript, which will be a direction of our further work.

Comment 45: Figure 6, C1qhi Mac only are few cells/events, and interactions (or cells?) seem to be gone in the Prdm1-floxed mice. Is that true? Does it make sense to perform cell-chat analysis on so few cells?

Response 45: We have now added KCs to the cell-chat analysis, and this cluster was belonged to C1qhi KCs. We have revised the analysis of corresponding parts in our manuscript (page 20-21; line 408-428):

“According to the transcriptional profile, liver macrophages further clustered and were labeled as “_Ly6c2_hi”; “_Cxcl2_hi”; “_Ear2_hi” MDMs, and “_Mrc1_hi”; “_C1q_hi” KCs (Figure 6A, Supplemental Figure 9, A-E). Increased proportion of MDMs and KCs was observed in *Prdm1ΔNcr1* cells (Supplemental Figure 9B). Within MDMs clusters, _Ly6c2_hi MDMs mainly compose of *Prdm1*^{+/+} cells, while *Prdm1ΔNcr1* cells concentrated in _Cxcl2_hi cluster (Figure 6C). The scRNA-seq data reveal that following *Prdm1* knockout in NKp46⁺ cells, there is a

decrease in the proportion of KCs within the macrophage population, while the proportion of MDMs increases (Figure 6D). CX3CR1, a chemokine receptor, is extensively utilized to distinguish KCs and MDMs within macrophages. Cells expressing CX3CR1 are identified as MDMs, whereas those without CX3CR1 expression are categorized as KCs (56). Employing flow cytometry and leveraging CX3CR1 expression, we assessed the ratios of KCs and MDMs. However, diverging from the scRNA-seq findings, flow cytometry indicates that post-Prdm1 knockout in group 1 ILCs, there is a minor increase in the proportion of KCs within the total liver macrophages, and a decrease in the proportion of MDMs (Figure 6D; Supplemental Figure 9B). This discrepancy could stem from the different bases of classification: scRNA-seq defines KCs based on gene expression profiles, whereas flow cytometry differentiates between KCs and MDMs using the single surface marker, CX3CR1. Analysis of the macrophage subsets identified by scRNA-seq reveals that, while MDM clusters generally show high CX3CR1 expression, there exists a subset within MDMs, labeled *Mrc1hi*, that also exhibits high levels of CX3CR1 (Supplemental Figure 9C). Consequently, if flow cytometry solely employs CX3CR1 for differentiating between KCs and MDMs, it could result in disparities when compared to scRNA-seq outcomes.”.

Comment 46: Figure 6, C - Here the interactions of both Mac+ILC1 and Mac+NK are shown together. It would be interesting to separate this analysis (also Suppl. Fig 9A-B) into comparisons of Mac+ILC1 vs Mac1+NK from WT or Prdm1fl Ncr1 mice.

Response 46: As request, we re-analyzed this part in each genotype, which was showed in the Supplemental Figure 10. These data have now been described in (page 22; line 445-447).

“The reduction of interaction mostly occurred in the cross-talk of ILC1-MDM and ILC1-KC, whereas no difference was observed in cNK-MDM and cNK-KC interaction (Supplemental Figure 10, A-H)”

Comment 47: Supplementary Figure 9, A, B - Is this analysis using WT and Prdm1fl Ncr1cre dataset together?

Response 47: Yes, we used WT and *Prdm1ΔNcr1* data together. As the request above, we separate this analysis from WT or *Prdm1ΔNcr1* Ncr1 mice. These data have now been described in (page 22; line 445-460):

“The reduction of interaction mostly occurred in the cross-talk of ILC1-MDM and ILC1-KC, whereas no difference was observed in cNK-MDM and cNK-KC interaction (Supplemental Figure 10, A-H). A reduction in the interaction of ligand-receptor, such as Mif-CD74, Cxcl16-Cxcr6, and Cxcl10-Cxcr3 was observed in *Prdm1ΔNcr1* mice compared to *Prdm1+/+* mice (Supplemental Figure 11). Compared to *Prdm1+/+* mice, the information flow of CXCL and MIF pathways significantly decreased in *Prdm1ΔNcr1* mice (Figure 6, H and I; Supplemental Figure 10, B, D, F, and H). These pathways play a crucial role in facilitating macrophage migration. The CXCL signaling was sent from *_Ly6c2_hi_Cxcl2_hi* MDMs and *_C1q_hi* KC, targeting all ILC1 clusters and *_Cxcr3_hi* cNK cell clusters (Figure 6J). Of note, although the population of *_Cxcl2_hi* macrophage primarily comprised cells from *Prdm1ΔNcr1* mice, the interaction within the CXCL pathway between macrophages and group 1 ILCs was obviously less than *Prdm1+/+* sample (Figure 6J). These changes could be linked to a decreased population of ILC1s and *_Cxcr3_hi* cNK cell cluster in *Prdm1ΔNcr1* mice, implying that the homeostasis of *_Cxcl2_hi* macrophages required sufficient signals from cNK cells and ILC1s. The impaired CXCLCXCR interactions might subsequently lead to reduced recruitment and activation of group 1 ILCs and macrophages within the tumor microenvironment.”.

Comment 48: Figure 7, A-C -What is the consequence/interpretation of reduced Mitotracker staining? Any metabolic assays performed? The definition of NK cell "exhaustion" is unclear, is reduced IFNg enough for that? Is the concept of NK cell

exhaustion clearly established? Only shortly touched upon in the discussion, the rationale for suggesting an exhausted phenotype, should be explained.

Response 48: MitoTracker was used to assess the mitochondrial mass. The reduced staining indicated compromised mitochondria function, which associated with mitochondrial fragmentation.

We believe that the exhaustion of NK cells is not as well-established a concept as it is for T cells. The purpose of detecting mitochondria in this study is to provide evidence for the relationship between *Prdm1* and the exhaustion of NK cells. In the discussion section, we have added the following content (page 26; line 529-543):

“Mitochondria are pivotal organelles crucial for cellular metabolism. Disruptions in mitochondrial function have been linked to T Cell exhaustion, attributed to glycolytic reprogramming (66). Similarly, mitochondrial fragmentation has been closely associated with NK cell exhaustion (67).

However, the concept of NK cell exhaustion isn't as firmly established as it is for T cells. Exhausted NK cells should primarily exhibit diminished functions. This is characterized by a diminished ability to destroy tumor cells, a reduced capability to activate other components of the immune system, and compromised proliferation and survival rates. Additionally, this reduced functionality is associated with a decline in the expression of molecules responsible for cytotoxic activity, lower production of IFN- γ , and metabolic disturbances that may arise from mitochondrial dysfunction. While our current data is not sufficient to definitively classify these cells as exhausted NK cells, it supports that a subpopulation, referred to *_Junb_hi* cluster, demonstrates an exhaustion-like phenotype. The significant increase in this cell population following *Prdm1* knockout in NK cells may potentially be one of the reasons why *Prdm1* Δ *Ncr1* mice lose their tumor-killing capacity. Whether the excessive expression of JunB in NK cells is also a contributing factor to their exhaustion, similar to T cells(65), requires further investigation.”.

Comment 49: Figure 7, x-axis labelling (MFI) of histograms is not correct. Do bar graphs and FACS plots show the same data? Does the number in the FACS plots indicate the MFI? If so, the FACS plots do not show representative samples?

Response 48: We appreciate the valuable comments provided by the reviewer. In the revised Figure 7, the MFI values have been removed. Bar graphs now display summary data from FACS histograms.

A representative sample close to the group's mean value was chosen for display in the histograms.

Comment 50: Figure 7, D - How are these data different from Figure 2H? Why is it now called "exhaustion", but not in 2H? Is the detected IFN γ only driven by ex vivo stimulation with IL12/IL18? As above, a "standard" 4h assay should also be provided to allow better interpretation of potential differences. In the introduction, the authors cite the Ducimetiere study (Ref 5) highlighting "the primary function of ILC1 in suppressing the seeding of metastatic tumor cells in liver tissue". Thus, it would be interesting to test Ifng production by liver ILC1 and NK cells ex vivo at early time points of tumor inoculation.

Response 50: Tumors grow and proliferate within tissues, constituting one of the major causes of lymphocyte exhaustion. This part of the current study aims to investigate whether *Prdm1* aids NK cells or ILC1 in resisting the exhaustion induced by malignant tumors. Specifically, we seek to ascertain whether the absence of *Prdm1* renders NK cells or ILC1 more susceptible to exhaustion within the tumor microenvironment. Therefore, we will

consider the capacity to secrete IFN- γ upon IL-12/IL-18 stimulation as one indicative aspect of exhaustion. It's crucial to emphasize that this assessment serves as only one piece of evidence, not the sole determinant. Overnight stimulation is a conventional method for studying NK cells and has been widely used across different laboratories, including our lab (e.g. Bream et al., *Blood*, 2003; Yu et al., *Immunity*, 2006; Wang et al., *J Clin Invest*, 2018). It's essential to clarify that our approach does not involve stimulating with tumor cells to evaluate the secretion capacity of IFN- γ by NK cells or ILC1.

Reviewer 2 (Public Review):

Summary:

This study offers a significant advancement in understanding liver innate lymphoid cell (ILC) biology by elucidating the role of the transcription factor Prdm1. It shows that Prdm1 is crucial in maintaining the balance between conventional natural killer (cNK) cells and ILC1s in the liver, with knockout models revealing a vital role in cancer defense mechanisms. Despite not affecting direct cytotoxicity, Prdm1 deficiency leads to increased cancer metastasis and reduced secretion of key molecules like IFN- γ , pointing to its importance in immune regulation. The use of single-cell RNA sequencing further underscores Prdm1's role in cellular communication within the liver's immune milieu. This study is a robust contribution to the field, providing insights that could inform new immunotherapy approaches for liver cancer.

Strengths:

The study's strength lies in its comprehensive approach, combining the specificity of Prdm1 conditional deletion in Ncr1-cre mice with integrative omics analyses and cutting-edge cytometry to delineate Prdm1's role in liver Type 1 ILC biology and its functional implications in tumor immunity. This multifaceted strategy not only clarifies Prdm1's influence on ILC composition and maturation but also conveys potential therapeutic insights for liver cancer immunotherapy.

We sincerely appreciate your interest and critical assessment of our manuscript. We have carefully read your comments and suggestions, and I am truly grateful for your expert guidance. We have worked on addressing each of your concerns and comments, and below we provide a point-to-point response. Please find the detailed responses below:

Weakness

Comment 1: A notable weakness of the study is the limited scope of in vivo disease models, primarily relying on the B16F10 melanoma model, which may not fully capture the complex behavior of Type 1 ILCs across diverse cancer types. Furthermore, the absence of direct human data, such as the effects of PRDM1 deletion in human NK cells or stem cells during their differentiation into NK and ILC1, leaves a gap in translating these findings to clinical settings.

Response 1: We appreciate the reviewer for raising these important points, which we see as a unique opportunity for future work to transform our understanding of Prdm1 and its targets as opposed to a weakness of the present study.

In our revised manuscript, we have discussed these limitations of our study (page 29; line 602-609):

“While our findings underscore the importance of Prdm1 in liver cNK cells and ILC1s tumor immune surveillance, it does not be validated in human NK cells, whereas previous studies have found that PRDM1 might inhibit the proliferation and function of human NK cells (33,

73). Furthermore, we not provided an in-depth evaluation in multiple tumor models. Further research may provide deeper insight into the role of PRDM1 in the anti-tumor function of human NK cells, enabling a more direct investigation of its application in cancer therapies. Given its important role in preserving liver cNK cells and ILC1s functional heterogeneity, enhancing Prdm1 function in human NK cells could potentially be a strategy to promote NK cell-based immunotherapy for cancer.”.

Recommendations For The Authors:

(Introduction)

Comment 2: Reference 1 appears slightly misplaced. You might find the nomenclature discussion in Spits et al., Nature Reviews Immunology, 2013, more appropriate.

Response 2: We are really sorry for our inaccurate descriptions. According to Spits et al., (Spits et al., *Nature Reviews Immunology*, 2013) and other related studies, we have now adopted a more appropriate nomenclature as “Conventional NK cells” correspond to “cNK cells”, “Type 1 innate lymphoid cells” to “ILC1s”, and “Group 1 ILC” as the collective name of cNK and ILC1s.

The definition of these cells was described in the introduction (page 4, line 52-53; line58-62):

“Group 1 ILCs consist of cNK cells and ILC1s (1, 2), with distinct developmental trajectories and effect molecules (3).”, “In a state of homeostasis, liver group 1 ILCs (CD45+CD3-NK1.1+NKp46+) can be discriminated into cNK cells and ILC1s by the differential expression of CD49a and CD49b (2): cNK cells are marked by the expression of CD49b, while liver ILC1s exhibit a distinctive positivity for CD49a. Tumor Necrosis Factor Related Apoptosis Inducing Ligand (TRAIL) is also expressed on liver ILC1s, but not on cNK cells (10, 11).”.

We also describe cNK and ILC1 phenotypes in our scRNA-seq data, as shown in page 13; line 259-261:

“cNK cells expressed high levels of *Itga2* (CD49b) and *Eomes*, while ILC1s had high levels expression of *Itga1* (CD49a) and *Tnfrsf10* (Supplemental Figure 5, F and G).”.

Comment 3: It has come to my attention that Reference 9 has been retracted. I recommend removing this citation to maintain the integrity of your references (<https://doi.org/10.1182/blood.2023022801>).

Response 3: We thank the reviewer’s comment and we now have removed this citation.

Comment 4: For a more comprehensive context around reference 15, consider citing Thierry Walzer’s work (<https://rupress.org/jem/article/211/3/563/41636/T-bet-and-Eomes-instruct-the-development-of-two>)) which aligns closely with your discussion.

Response 4: We agree with the reviewer’s suggestion and have added this citation in our introduction (page 4; line 64-66):

“Liver environment facilitated T-bet expression in the early stage of NK cells development, which results in *Eomes* repression. The repression of T-bet is required for *Eomes*+ NK cells (17).”.

(Results)

Comment 5: The NK cell signature referenced in 32 has been questioned for its reliability as discussed by Cursons et al., CRI 2019 (<https://pubmed.ncbi.nlm.nih.gov/31088844/>).

Reanalysis of data in Figure 1 B/C and Supplementary Figure 1 with the refined NK cell signature from Curson's work would be advantageous.

Response 5: We thank the reviewer's comment. As requested, we reanalyzed our data using the refined NK cell signature from Cursons et al. (revised Figure 1 A-C; revised Supplemental Figure 1). Of note, the overall survival of liver cancer (LIHC) patients only reached statistics significance when compared high and low expression of refined PRDM1-NK signature with a median cutoff (Figure 1, A-C). The overall survival performed with quartile high and low expression of refined PRDM1-NK signature was moved to supplemental figure 1, G-I.

The original text is: "Examination of 363 liver hepatocellular carcinoma (LIHC) patient samples from The Cancer Genome Atlas (TCGA) revealed a positive correlation between the expression of NK cell-associated genes (*NCR1*, *NCR3*, *KLRB1*, *CD160*, and *PRF1*) (32) and *PRDM1* expression (Figure 1A). Patients with top and bottom quartiles of *NK-PRDM1* signature expression were chosen for survival analysis (Figure 1B). Notably, patients with the *_NK-PRDM1_hi* signature had better overall survival compared to the these with *NK-PRDM1_lo* signature (Figure 1C). Similar results were also found in skin cutaneous melanoma (SKCM, n=454) and lung adenocarcinoma (LUAD, n=497) patients (Supplemental Figure 1, A-F). These data suggested that *PRDM1* in NK cells might be essential for immune surveillance in some solid tumors, including liver cancer. These findings prompted us to investigate the impact and mechanism of *PRDM1* in NK cells and ILC1 within the context of liver cancer."

We have rewritten this part in our revised manuscript (page 7; line 119-132):

"Examination of 363 liver hepatocellular carcinoma (LIHC) patient samples from The Cancer Genome Atlas (TCGA) revealed a positive correlation between the expression of NK cell-associated genes (34) (*NCR1*, *KLRB1*, *CD160*, *PRF1*, etc.) and *PRDM1* expression (Figure 1A). The patients are ordered from highest to lowest based on the expression of NK-Prdm1 for survival analysis (Figure 1B). Notably, patients exhibiting higher levels of NK-PRDM1 expression (above the median) experienced better survival outcomes compared to those with lower levels of NK-PRDM1 expression (below the median) (Figure 1C). Similar results were also found in skin cutaneous melanoma (SKCM, n=454) and lung adenocarcinoma (LUAD, n=497) patients (Supplemental Figure 1, A-F). Patients within the highest quartile of NK-PRDM1 signature expression demonstrated enhanced overall survival, a result that achieved statistical significance in LUAD and SKCM patients (Supplemental Figure 1, G-I). These data suggested that *PRDM1* in NK cells might be essential for immune surveillance in solid tumors, including liver cancer, and prompted us to investigate the function and mechanism of *PRDM1* in NK cells and ILC1 within the context of liver cancer."

Comment 6: The origin of the Ncr1-cre mice utilised should be clarified; is this the line developed by Eric Vivier? (<https://www.pnas.org/doi/10.1073/pnas.1112064108>).

Response 6: We did not use the line developed by Eric Vivier, our *Ncr1-cre* mice was purchase from Shanghai Model Organism Center, Inc.. We described this in our method parts (page 29-30; line 612-614):

"*Prdm1^{fl/fl}* mice were purchased from The Jackson Laboratory. *Ncr1-iCre* and *B2m^{-/-}* mice were purchased from Shanghai Model Organisms Center, Inc.. Six- to twelve-week-old littermates were used for the experiment."

Comment 7: Considering the known reduction of Ncr1 expression in Ncr1-cre mice and its implications, it is recommended to repeat the B16F10 experiments with the correct control, Ncr1cre/+ Prdm1+/-.

Response 7: This is an excellent question, and it has been raised by another reviewer and comprehensively answered (Reviewer 1, Comment 1). The answer is below:

The expression of Cre and the insertion of loxP sequences both have the potential to influence gene expression. This is because the region where loxP is inserted may contain regulatory sequences for the gene of interest. Ncr1-Cre is a frequently used transgenic mouse model in our laboratory. In our prior research, we also had concerns about the possible impact of Cre on Nkp46 expression, which could lead to a decline in NK cell function. Therefore, in our previous studies focused on Smad4 expression in NK cells, we conducted similar experiments. In Figure 6 of our published paper in the Journal of Clinical Investigation (Wang et al., *J Clin Invest*, 2018), we compared Nkp46iCreTgfb2fl/flSmad4fl/WT with Nkp46-iCreTgfb2fl/flSmad4fl/fl. Although the primary purpose is to establish Smad4's independence from TGF- β , it also allows for a comparison between Smad4fl/fl and Smad4fl/WT in the presence of Cre. In the critical phenotype we assessed, Nkp46iCreTgfb2fl/flSmad4fl/fl (compared with Nkp46-iCreTgfb2fl/flSmad4fl/WT) exhibited the same phenotype as Nkp46-iCreSmad4fl/fl (compared with Nkp46WTsmad4fl/fl). This suggests that Cre's influence on NK cells may be within a reasonable and controllable range. Furthermore, in contrast to the decrease in Ncr1 expression caused by Cre, the reduction in the expression levels of genes targeted by Loxp knockout, such as Prdm1 in this study (Figure 1 E), is more significant. Therefore, with the current techniques and research methods, we believe that the data provided in this study can support the role of Prdm1 in NK cells.

Comment 8: The proportion of ILC1 in wild-type mouse livers is notably higher than standard references. Could you confirm whether liver perfusion was performed before analysis? This procedure was not clearly detailed in the methods section.

Response 8: We apologize that we did not provide enough detail regarding this point in our original method. We had performed the liver perfusion before analysis. This has now been clarified in the method section of the revised text (page 30-31; line 630-636):

“Mice were perfused with 1 $\times$ PBS by portal vein puncture before harvesting tissues. Liver and lung was digested with 0.05% collagenase II for 30 minutes and filtered through 70 μ m cell strainers, and mononuclear cells were isolated after subjected to density gradient using 30% and 70% percoll. Spleen were also removed and pressed through 70 μ m filterers to obtain splenocytes. Peripheral blood mononuclear cells were obtained from peripheral blood after lysis of red blood cells (Biolegend, 420301). Flushing femurs and mechanical disruption of inguinal lymph nodes were performed to obtain cells from bone marrow and lymph nodes.”.

The lymphocyte proportions in mice from different laboratories may exhibit slight variations, possibly due to genetic background disparities. To minimize the influence of genetic backgrounds, paired littermates were used in the current study, wherein one is Prdm1 WT and the other has the Prdm1 gene knocked out in NK cells.

Comment 9: There appears to be inconsistency in reference formatting; for instance, Ref 39 does not match the formatting of other references. A thorough review of your citation format is suggested.

Response 9: We apologize for the inadvertent errors and we reviewed the citation format.

Comment 10: The information in Figures 2B and C may be better suited to the supplementary section as it does not significantly contribute to the main text.

Response 10: We agree with the reviewer's suggestion and these are now moved to supplementary figures (Supplemental Figure 2).

Comment 11: The citation of reference 40 could be strengthened by including Sathe et al., 2014, which directly pertains to your findings (<https://www.nature.com/articles/ncomms5539>).

Response 11: We added the suggested reference.

Comment 12: Can the findings presented in Figure 2D/F be replicated using alternative models?

This would substantiate the versatility of your results.

Response 12: The current predominant in vivo tumor model for NK cells is primarily based on the use of B16F10 melanoma cells. These melanoma cells, with their low expression of MHC-I molecules, evade T cell-mediated immune surveillance, rendering them ideal targets for NK cells. Typically, this experimental melanoma metastasis assay involves tail vein injection, followed by nodules' detection in the lungs. To align with our investigation of liver-resident cNK and ILC1, we've introduced splenic injection (via the portal vein) and evaluated melanoma metastasis in the liver to reflect the anti-tumor capabilities of liver group 1 ILCs. We also explored subcutaneous tumor models, but we believe they may not effectively support Prdm1's role in cNK cells, particularly liver-resident NK cells and ILC1. While we've experimented with models using mouse liver tumor cells like Hepa 1-6, we found them less stable than B16F10 and less conducive to quantification. Should more suitable models or cells line emerge, we remain open to exploring them in future research.

Comment 13: The absence of in vitro killing assessments against B16F10 and YAC-1 leaves a gap in the NK cell characterisation which would be valuable to address.

Response 13: Isolating NK cells for ex vivo cytotoxicity assays typically requires stimulation with high concentrations of IL-2. Under such high IL-2 stimulation, many intracellular differences that contribute to difference in cytotoxicity, such as changes in transcription factors, are often masked. Another issue is that current ex vivo NK cell cytotoxicity assays often only isolate NK cells from the spleen. Liver-resident NK cells, on the other hand, are often limited in quantity and isolation methods, making it challenging to conduct ex vivo cytotoxicity assays effectively. If more sensitive detection methods become available, we will also incorporate ex vivo data into our future research endeavors.

Comment 14: The suggestion that NK cells produce IL-6 is indeed a bold one, and without additional validation through intracellular cytokine detection or ELISA, it may be prudent to omit these claims.

Response 14: We have checked the GSEA results, and found no valuable genes in IL-6 production.

Therefore, we have removed this figure.

Comment 15: The lack of fluorescence minus one (FMO) controls in Figure 3 and Supplementary

Figure 4 is noted; including these would enhance the validity of your gating strategies.

Response 15: As requested, we add the FMO controls in aforementioned figures.

Comment 16: There seems to be a minor mix-up in referring to Figure 4A in the scRNAseq results section, perhaps it was intended to refer to Figure 3A?

Response 16: We have corrected this part (line 247). We also double checked corrected the inaccuracies in the references to the figures. we apologize for the inadvertent errors.

Comment 17: The rich datasets generated from bulk and scRNAseq are commendable. However, I urge you to make these datasets publicly accessible with a GEO accession number.

Response 17: We appreciate the suggestion from the reviewer. We plan to upload our datasets when in the last version of our manuscript, which is also the request of the eLife policy.

Comment 18: Figure 4K is insightful, yet a similar analysis of the ILC1 cluster could provide a more rounded understanding.

Response 18: We thank the reviewer for the comments. We provide the similar analysis of ILC1s, as showing in revised Figure 5H.

Comment 19: The metabolic RNA signatures featured in Supplementary Figure 6 are intriguing and warrant further validation, perhaps through Seahorse analysis. Such validation could merit their inclusion in the main figures.

Response 19: This is a very good suggestion. Currently, our data offer only limited indications in this context. We have chosen to validate some aspects of Prdm1's influence on cytotoxicity molecules. As for Prdm1's impact on other aspects of NK cells, such as metabolic functions, we may explore further in future research. Additionally, we hope that by publishing our research findings, laboratories worldwide can draw insights for their own studies and conduct relevant research based on this data.

Comment 20: It is difficult to discern whether the cells depicted in Figure 7D are truly tumorinfiltrating ILC1 or NK cells that have adopted ILC1-like characteristics. Intravenous injection of CD45-PE could clarify this distinction, and if they are the latter, it may be more appropriate to refer to them as ILC1-like cells.

Response 20: We completely agree with the reviewer's suggestion that "tumor-infiltrating lymphocytes" may not be accurate for the current experiment. Therefore, in the revised manuscript, we have changed it to "liver cNK or ILC1 from tumor-bearing livers."

<https://doi.org/10.7554/eLife.92948.2.sa3>